PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^K}[T]$, II: EXACT VARIANCES VIA QUADRATIC TOWERS AND SUPERSINGULAR ARTIN–SCHREIER CURVES

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Abstract. Let $q = 2^k$ and let $\mathcal{I}_f = \{f^2 + s : \deg s \le d\}$ be the Legendre interval of a monic $f \in \mathbb{F}_q[t]$ of degree d. Chen's Open Problem 8.2 asks why the variance of the prime-power count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ deviates by a factor as large as $q^2/3$ from the Keating–Rudnick prediction. We resolve the case $d = 4$ completely and unconditionally: using the quadratic tower $\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4} \subset \mathbb{F}_{q^8}$ we obtain a closed formula for Ψ on every class of the Hayes group $\mathcal{G}_3$, from which $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$, $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$, the full structure of the quadratic-twist correlations, and an elementary proof of Carlitz's cubic evaluation over $\mathbb{F}_{2^{8k}}$ follow. For $d = 6$ we prove that the Legendre profile takes exactly three values governed by $[c_2 = 0]$ and $\mathrm{Tr}(c_4/c_2^2)$, reducing the variance to three exponential sums A_6, X, Y on $U = \{\alpha \in \mathbb{F}_{q^{12}} : \mathrm{Tr}\,\alpha = \mathrm{Tr}\,\alpha^3 = \mathrm{Tr}\,\alpha^5 = 0\}$. We show that $A_6 - q^9$ and X are sums of Frobenius traces of supersingular Artin–Schreier curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ and reduce them to $\mathbb{F}_2$-quadratic forms whose radicals are kernels of explicit linearised polynomials; the pure cubic and pure quintic parts are evaluated for all k, and for odd k the mixed part is evaluated completely: it is governed by the factorisation type of the Gold-type quintic $v^5 + v + c$ over $\mathbb{F}_q$, a $\mathbb{Z}/4$-condition on the Frobenius, and the signs are fixed through the L-polynomials of the curves. This proves $A_6 = q^9 + (q-1)q^5$ for all odd k. The conjectured closed forms in the remaining cases ($A_6 = q^9 - (q-1)q^5$ for even k, and $X = (-1)^{k+1}q^6 + q^5$ for all k) are verified exactly for $q \le 128$ by an algorithm independent of the explicit formula; for X we determine the structure of the corresponding order-4 character sums (whose Frobenius classes run through $S_6 \cong Sp_4(\mathbb{F}_2)$ with exact polynomial frequencies) and reduce the closed form to an explicit conjecture. The third constant $Y = q^3 m_k$ is a twisted Frobenius trace on a fixed nine-dimensional complete intersection $X = V(e_1, e_3, e_5) \subset \mathbb{A}^{12}_{\mathbb{F}_2}$ (geometrically integral, normal, with explicit two-dimensional singular locus) and is not a character sum; we discuss its weight and rank. Consequently the Legendre variance at $d = 6$ obeys $\mathrm{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$, not q^4. A spectral-folding analysis of the pair $\mathcal{G}^2 \subset \mathcal{G}$ identifies the amplification as the coherence of the Fourier coefficients of Ψ along the fibres of $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$, which are cosets of the 2-torsion $\mathcal{G}[2]$; it is produced by the identity class and its translation orbit, whose size is decided by Lucas' theorem. Exact computations up to $d = 10$ show that the amplification is not universal: at $d = 8$ the Legendre variance is only about 2.3 times the full-group variance for $q \le 8$.

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1. Introduction

Let $q = 2^k$, $d \geq 2$, $n = 2d$ and $\ell = d - 1$. For a monic $f \in \mathbb{F}_q[t]$ of degree d the *Legendre interval* is

$$\mathcal{I}_f = \{f^2 + s : \ s \in \mathbb{F}_q[t], \ \deg s \leq d\}, \qquad |\mathcal{I}_f| = q^{d+1},$$

and $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ counts prime powers in it with the von Mangoldt weight. In odd characteristic the sets $\mathcal{I}_f$ are ordinary short intervals and the variance of Ψ is governed by the Keating–Rudnick theorem [15], which predicts $\mathrm{Var}(\Psi) \sim (d-2)q^{d+1}$ as $q \to \infty$. In characteristic 2 the squaring map on the Hayes group is a homomorphism, the Legendre intervals are the fibres over the subgroup of squares, and Chen [5] observed (Table 4 of *op. cit.*) that the variance over Legendre intervals exceeds the full-group variance by factors $4/3$, $16/3$, $64/3$ for $q = 2, 4, 8$ at $d = 4$ and asked for an explanation (Open Problem 8.2).

This paper is the second part of our study. Part I established the dictionary between Legendre intervals and quadratic characters of the Hayes group and verified, by exact computation for $q \leq 8$, the identities $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$ and $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$. Here we prove them for all $q = 2^k$, and we carry the analysis to $d = 6$, where a genuinely new phenomenon appears: two of the three constants that control the Legendre variance are elementary while the third is a Frobenius trace that is not a character sum.

Main results. All variances below are second moments about the global mean q^{d+1} (this is the quantity tabulated in [5]); see Section 2.

- $d = 4$ **(Section 3).** Theorem 3.2 gives Ψ on every class of the Hayes group $\mathcal{G}_3$ in closed form. Consequently (Theorems 3.4, 3.11, 3.12) for every $q = 2^k$

$$\frac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1), \qquad \frac{\mathrm{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}, \qquad \frac{\mathrm{Var}_{\mathcal{G}^2}}{\mathrm{Var}_{\mathcal{G}}} = \frac{q^2}{3},$$

the deviation $\Psi - q^5$ is supported on the subgroup $\{c_1 = 0\}$, the identity class carries the deficit $-(q^4 - q^3)$, and the quadratic-twist correlations $\rho(\varepsilon)$ equal 1 on the $q-1$ characters of conductor u^2 and $(q-3)/(3(q-1))$ on all others when k is odd.

- $d = 6$, **structure (Section 4).** Theorem 4.2: on the q^2 Legendre classes $(0, c_2, 0, c_4, 0)$ the function Ψ takes exactly three values,

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q}[c_2 = 0] + \frac{Y}{q}[c_2 \neq 0]\,\chi\Big(\frac{c_4}{c_2^2}\Big),$$

  where $A_6 = |U|$, $X = \sum_U \chi(e_2)$, $Y = \sum_U \chi(e_4)$ and $U = \{\alpha \in \mathbb{F}_{q^{12}} : \operatorname{Tr}\alpha = \operatorname{Tr}\alpha^3 = \operatorname{Tr}\alpha^5 = 0\}$. The proof uses only the $\mathbb{F}_q$-translation and scaling symmetries of U.
- $d = 6$, **the constants (Sections 5–6).** Theorem 5.3 reduces A_6, X, Y to explicit finite sums over $\mathbb{F}_{q^6}$. Theorem 6.1 expresses $A_6 - q^9$ and X through the Frobenius traces of the supersingular curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$, equivalently through $\mathbb{F}_2$-quadratic forms on $\mathbb{F}_{q^{12}}$ with explicit radicals. Lemmas 6.2 and 6.4 evaluate the pure cubic and pure quintic parts for all k. For odd k, Theorem 6.12 evaluates the mixed part: $W(\lambda_3, \lambda_5) = 2q^7$ or 0 according as the Frobenius permutes the roots of the Gold-type quintic $v^5 + v + c$, $c = (\lambda_5^3/\lambda_3^5)^{1/2}$, as a 4-cycle or not; with the exact count of 4-cycles (Proposition 6.8) this *proves* $A_6 = q^9 + (q-1)q^5$ for all odd k (Corollary 6.13). What remains conjectural is Conjecture 6.5 in the other cases, $A_6 = q^9 - (q-1)q^5$ for even k and $X = (-1)^{k+1}q^6 + q^5$ for all k, verified exactly for $q \le 128$ (Table 1); for odd k the sums behind X are analysed in §6.7.
- **Scaling law and Y (Section 7).** Theorem 7.1 gives $\operatorname{Var}_{\mathcal{G}^2}(\Psi)/q^7$ at $d = 6$ in closed form in terms of (q, k, m_k), $m_k = Y/q^3$; it implies $\operatorname{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ (Table 4) and $\limsup \operatorname{Var}_{\mathcal{G}^2}/q^9 = 1$ provided $m_k = o(q^{5/2})$. We identify Y as the trace of a twisted Frobenius σF^k on the cohomology of a fixed nine-dimensional complete intersection $X \subset \mathbb{A}^{12}_{\mathbb{F}_2}$ with an Artin–Schreier sheaf (Propositions 7.4–7.5, Theorem 7.7) and show that the data $m_k = 13, -1, -35, 47, 733$ $(k \le 5)$ exclude a pure weight-3 object of rank ≤ 5. Y is the difference of the twisted point counts of two eight-dimensional complete intersections (Proposition 7.9), and $C_6 = 1$ is equivalent to a Lefschetz-type weight bound for them.
- **Spectral folding (Section 8).** For a subgroup H of a finite abelian group, the variance over H is the energy of the spectrum folded along the fibres of $\widehat{G} \to \widehat{H}$, and it exceeds the full variance by at most $[G : H]$, with equality iff the function is supported on H. For the Hayes group $[\mathcal{G} : \mathcal{G}^2] = |\mathcal{G}[2]| = q^{d/2}$, which is why the phenomenon is confined to characteristic 2. At $d = 4$ all fibre sums are computed exactly (Theorem 8.2); at $d = 6$ the Legendre variance is $[(A_6 - q^9)^2 + (q-1)X^2]/q^4 + q^3(q-1)m_k^2$ exactly, the coherent part being A_6 and X and the incoherent part Y.
- **General even d (Section 9).** The translation orbit K_d of the identity class has order 1 or q according as d is a power of 2 or not (Lucas), and Ψ is constant on it. Exact variances for $d \le 10$ show that the q^2-amplification is special to $d = 4, 6$: at $d = 8$ the ratio of the Legendre to the full-group variance is ≈ 2.3 for $q = 4, 8$, refuting both the $q^{2(d/2-1)}$ law of [5] and a universal q^2 law.

Methods. Two ideas carry the paper. The first is the *quadratic tower*: for $n = 2d$ the extension $\mathbb{F}_{q^n}/\mathbb{F}_{q^{n/2}}$ is quadratic, so every $\alpha \in \mathbb{F}_{q^n}$ is a root of $t^2 + at + n$ with $(a, n) \in \mathbb{F}^2_{q^{n/2}}$, and the elementary symmetric functions of the n conjugates of α become explicit polynomials in traces of (a, n). Counting pairs (a, n) with the irreducibility condition $\operatorname{Tr}_{q^{n/2}/2}(n/a^2) = 1$ turns the prime-counting problem into linear algebra over $\mathbb{F}_q$ and $\mathbb{F}_2$. The second is the use of the *symmetries of* $U_d = \{p_j = 0,\ j \text{ odd} \le \ell\}$: U_d is stable under $\alpha \mapsto \alpha + c$ $(c \in \mathbb{F}_q)$ and $\alpha \mapsto s\alpha$ $(s \in \mathbb{F}_q^\times)$, and the elementary symmetric functions transform by binomial coefficients mod 2; this alone forces the shape of the Legendre profile.

Reproducibility. All statements labelled "verified" were checked with exact integer arithmetic. The scripts that produce every number in the paper (brute force, the $d = 4$ formula, the Hayes FFT driver, the tower identities, the quadratic-form algorithm and its fine-structure variants,

the L-polynomial check, the figures) are in the `code/` directory of the repository and are named at the places where they are used. The Hayes-group FFT of [5] was used as an independent check for $q \le 32$ at $d = 6$ and $q \le 128$ at $d = 4$.

2. Preliminaries

2.1. **Hayes group, fibres, Legendre intervals.** Let $\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1 u + \cdots + c_\ell u^\ell\} \subset (\mathbb{F}_q[u]/u^{\ell+1})^\times$, $|\mathcal{G}| = q^\ell$. A monic $F = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ maps to $F^*(u) = u^n F(1/u) \bmod u^{\ell+1}$, i.e. to its top ℓ coefficients $(c_1, \dots, c_\ell) = (a_{n-1}, \dots, a_{n-\ell})$; the map is multiplicative and each fibre has $q^{n-\ell} = q^{d+1}$ elements.

Lemma 2.1. *The Legendre interval $\mathcal{I}_f$ is the fibre over the class of f^2, and the classes of squares form the subgroup $\mathcal{G}^2 = \{g^2\}$, of index q^m with $m = \#\{j \le \ell : \ j \text{ odd}\} = \lfloor d/2 \rfloor$. Hence $\Psi(f) = \Psi(g)$ depends only on $g = [f^2] \in \mathcal{G}^2$.*

Proof. $f^2 + s$ has vanishing coefficients in all odd degrees $> d$, and $(1 + \sum c_j u^j)^2 = 1 + \sum c_j^2 u^{2j}$. The fibre of $[f^2]$ is $\{f^2 + s : \deg s \le d\}$ since the top $\ell = d - 1$ coefficients of $f^2 + s$ are those of f^2. □

2.2. **Ψ as a root count.** For $\alpha \in \mathbb{F}_{q^n}$ write $\operatorname{charpoly}(\alpha) = \prod_{i<n}(t - \alpha^{q^i})$. If $F = P^j$ with $\deg P = n/j$, exactly $n/j = \Lambda(F)$ elements α have $\operatorname{charpoly}(\alpha) = F$. Hence

$$\Psi(g) = \sum_{F \in \text{fibre}(g)} \Lambda(F) = \#\{\alpha \in \mathbb{F}_{q^n} : \ (e_1, \dots, e_\ell)(\alpha) = g\}, \tag{1}$$

where $e_j(\alpha)$ denotes the j-th elementary symmetric function of the n conjugates of α. In particular $\sum_g \Psi(g) = q^n$ and the mean over $\mathcal{G}$ is q^{d+1}.

2.3. **Variances and quadratic twists.** Put $\varphi = \Psi - q^{d+1}$ and

$$\operatorname{Var}_{\mathcal{G}}(\Psi) = \frac{1}{|\mathcal{G}|}\sum_{g \in \mathcal{G}} \varphi(g)^2, \qquad \operatorname{Var}_{\mathcal{G}^2}(\Psi) = \frac{1}{|\mathcal{G}^2|}\sum_{g \in \mathcal{G}^2} \varphi(g)^2.$$

For a character χ of $\mathcal{G}$ let $\psi_n(\chi) = \sum_{\deg F = n} \Lambda(F)\chi(F)$. Then $\varphi(g) = |\mathcal{G}|^{-1}\sum_{\chi \ne 1} \psi_n(\chi)\bar{\chi}(g)$, and for ε a character trivial on $\mathcal{G}^2$,

$$\rho(\varepsilon) := \frac{\sum_\chi \psi_n(\chi)\overline{\psi_n(\chi\varepsilon)}}{\sum_{\chi \ne 1} |\psi_n(\chi)|^2} = \frac{\sum_g \varphi(g)^2 \varepsilon(g)}{\sum_g \varphi(g)^2}, \qquad \frac{\operatorname{Var}_{\mathcal{G}^2}}{\operatorname{Var}_{\mathcal{G}}} = 1 + \sum_{\varepsilon \ne 1,\ \varepsilon|_{\mathcal{G}^2} = 1} \rho(\varepsilon). \tag{2}$$

For $\chi \ne 1$ the L-function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\le \ell - 1 = d - 2$ with inverse roots of absolute value $\sqrt{q}$, and $\psi_n(\chi) = -\sum_j \gamma_j^n$. Thus $\operatorname{Var}_{\mathcal{G}}(\Psi)/q^{d+1} = |\mathcal{G}|^{-1}\sum_{\chi \ne 1} |\operatorname{Tr}\Theta_\chi^n|^2$ with $\Theta_\chi \in U(d-2)$ the normalised Frobenius; the Keating–Rudnick prediction is the Haar average $d - 2$.

2.4. **Quadratic characters are Artin–Schreier characters.**

Proposition 2.2. *Let $p_j(g)$ be the j-th power sum of the "roots" of $g \in \mathcal{G}_\ell$, defined by Newton's identities (in characteristic 2: $p_j = \sum_{i<j} c_i p_{j-i} + [j \text{ odd}] c_j$). Then $p_j(gh) = p_j(g) + p_j(h)$, $p_{2j} = p_j^2$, and $g \mapsto (p_j(g))_{j \text{ odd} \le \ell}$ induces an isomorphism $\mathcal{G}/\mathcal{G}^2 \cong \mathbb{F}_q^m$. The q^m quadratic characters are*

$$\varepsilon_\lambda(g) = (-1)^{\operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_2} \sum_{j \text{ odd}} \lambda_j p_j(g)}, \qquad \lambda \in \mathbb{F}_q^m,$$

and on a root α of a polynomial in the class g, $\varepsilon_\lambda(\alpha) = \psi_0(\sum_j \lambda_j \alpha^j)$ with ψ_0 the canonical additive character of $\mathbb{F}_{q^n}$. In particular ε_λ is primitive (does not factor through $\mathcal{G}_{\ell-1}$) if and only if $\lambda_\ell \ne 0$, which requires ℓ odd, i.e. d even.

Proof. Power sums are additive on products because roots are concatenated; $p_j(g^2) = 2p_j(g) = 0$ for odd j and $p_{2j} = p_j^2$ in characteristic 2. The map is onto $\mathbb{F}_q^m$ because $p_j = c_j + (\text{terms in } c_{<j})$ lets one solve for the odd c_j; its kernel has order $q^{\ell-m} = |\mathcal{G}^2|$ and contains $\mathcal{G}^2$. Finally $p_j(\operatorname{charpoly}\alpha) = \operatorname{Tr}_{q^n/q}(\alpha^j)$ and $\operatorname{Tr}_{q/2} \circ \operatorname{Tr}_{q^n/q} = \operatorname{Tr}_{q^n/2}$. □

We write $\chi(x) = (-1)^{\mathrm{Tr}_{q/2}(x)}$ for $x \in \mathbb{F}_q$ and $W = \{w \in \mathbb{F}_q : \mathrm{Tr}_{q/2}\, w = 1\}$. Throughout, $\varepsilon_0 := (-1)^{k+1}$.

3. The case $d = 4$: unconditional theorems

Here $n = 8$, $\ell = 3$, $g = (c_1, c_2, c_3) = (a_7, a_6, a_5)$, $\mathcal{G}^2 = \{(0, c, 0)\}$ has order q, and $\mathcal{G}_0 := \{c_1 = 0\} \cong (\mathbb{F}_q^2, +)$ is a subgroup of order q^2 containing $\mathcal{G}^2$.

3.1. The quadratic tower.

Lemma 3.1. *For $\alpha \in \mathbb{F}_{q^8}$ put $a = \alpha + \alpha^{q^4}$, $n = \alpha^{1+q^4} \in \mathbb{F}_{q^4}$ and $\mathrm{Tr} = \mathrm{Tr}_{q^4/q}$. Then*

$$e_1(\alpha) = \mathrm{Tr}\, a, \qquad e_2(\alpha) = e_2(a) + \mathrm{Tr}\, n, \qquad e_3(\alpha) = e_3(a) + \mathrm{Tr}\, a\ \mathrm{Tr}\, n + \mathrm{Tr}(an),$$

where $e_j(a)$ are the elementary symmetric functions of the four conjugates of a. A pair (a, n) has exactly two preimages α if $t^2 + at + n$ is irreducible over $\mathbb{F}_{q^4}$ (equivalently $a \neq 0$ and $\mathrm{Tr}_{q^4/2}(n/a^2) = 1$), one preimage if $a = 0$ ($\alpha \in \mathbb{F}_{q^4}$, $n = \alpha^2$), and none otherwise.

Proof. The eight conjugates of α are the roots of $\prod_{i<4}(t^2 + a^{q^i} t + n^{q^i})$; expand. If $t^2 + at + n$ is reducible with $a \neq 0$ its roots $\beta \in \mathbb{F}_{q^4}$ satisfy $\beta + \beta^{q^4} = 0 \neq a$. □

Two facts are used repeatedly: (i) $\mathrm{Tr}_{q^8/2}$ vanishes on $\mathbb{F}_{q^4}$, $\mathrm{Tr}_{q^4/q}$ vanishes on $\mathbb{F}_{q^2}$ and on $\mathbb{F}_q$; (ii) for $a \in \mathbb{F}_{q^4}$ the $\mathbb{F}_q$-linear forms $\mathrm{Tr}(n)$, $\mathrm{Tr}(an)$ are independent iff $a \notin \mathbb{F}_q$, and the $\mathbb{F}_2$-form $\mathrm{Tr}_{q^4/2}(a^{-2}n)$ lies in the $\mathbb{F}_2$-span of $\{\mathrm{Tr}_{q^4/2}(cn) : c \in \mathbb{F}_q + \mathbb{F}_q a\}$ iff $a \in \mathbb{F}_{q^2}$.

3.2. The closed formula.

Theorem 3.2. *For every $q = 2^k$,*

$$\Psi(c_1, c_2, c_3) = q^5 \qquad (c_1 \neq 0),$$
$$\Psi(0, c_2, c_3) = q^5 + q^3[c_3 = 0] + 2q^3\, I(c_2, c_3) - 2q^2\, K(c_2, c_3),$$

where $I(c_2, c_3) = [\, c_2 c_3 \neq 0$ and $\mathrm{Tr}_{q/2}(c_2^3/c_3^2) = 1\,]$ and

$$K(c_2, c_3) = \sum_{w \in W} \sum_{v \in \mathbb{F}_q} \chi\Big(\frac{c_2 v^2}{w} + \frac{\sqrt{c_2}\, v}{w} + \frac{c_3 v^3}{w^2}\Big).$$

Proof. We count pairs (a, n) as in Lemma 3.1 with $(e_1, e_2, e_3) = (c_1, c_2, c_3)$, according to the degree of a over $\mathbb{F}_q$.

Case I: $a \notin \mathbb{F}_{q^2}$. We need $\mathrm{Tr}\, a = c_1$ and then two independent $\mathbb{F}_q$-linear conditions on n, $\mathrm{Tr}\, n = c_2 + e_2(a)$ and $\mathrm{Tr}(an) = c_3 + e_3(a) + c_1(c_2 + e_2(a))$, cutting out an affine subspace of size q^2; the irreducibility condition is a non-constant affine $\mathbb{F}_2$-functional on it by fact (ii), so exactly $q^2/2$ values of n qualify, each with two roots. The number of a is $q^3 - q^2[c_1 = 0]$ (all $a \in \mathbb{F}_{q^2}$ have $\mathrm{Tr}\, a = 0$). Contribution: $q^5 - q^4[c_1 = 0]$.

Case II: $a \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$. Then $\mathrm{Tr}\, a = 0$, so $c_1 = 0$. Let $T = a + a^q \in \mathbb{F}_q^\times$, $N = a^{1+q}$; the conjugates are a, a^q, a, a^q, so $e_2(a) = T^2$ and $e_3(a) = 0$. The conditions are $\mathrm{Tr}\, n = c_2 + T^2$, $\mathrm{Tr}(an) = c_3$ (q^2 solutions). Writing $a^{-2} = x + ya$ with $x = (N + T^2)/N^2$, $y = T/N^2$ (from $a^2 = Ta + N$), the irreducibility functional equals $\mathrm{Tr}_{q/2}(x(c_2 + T^2) + yc_3)$ on the solution set, and $\mathrm{Tr}_{q/2}(xT^2) = \mathrm{Tr}_{q/2}(z + z^2) = 0$ with $z = T^2/N$. Hence the contribution is $2q^2$ times the number of a with

$$\tau(a) := \mathrm{Tr}_{q/2}\Big(\frac{(N + T^2)c_2 + Tc_3}{N^2}\Big) = 1.$$

Such a correspond two-to-one to irreducible $t^2 + Tt + N$, i.e. to (T, N) with $\mathrm{Tr}_{q/2}(N/T^2) = 1$. Substituting $N = T^2 w$ ($w \in W$), $v = 1/T$ and using $\mathrm{Tr}_{q/2}(c_2 v^2/w^2) = \mathrm{Tr}_{q/2}(\sqrt{c_2}\, v/w)$ gives contribution $4q^2 M$ with $M = \#\{(v, w) \in \mathbb{F}_q^\times \times W : \tau = 1\} = q^2/4 - K/2$, i.e. $q^4 - 2q^2 K$.

Case III: $a \in \mathbb{F}_q^\times$. Then $c_1 = 0$, $e_2(a) = e_3(a) = 0$, the conditions read $\mathrm{Tr}\, n = c_2$, $a\,\mathrm{Tr}\, n = c_3$, and irreducibility is $\mathrm{Tr}_{q/2}(a^{-2}c_2) = 1$. This forces $c_2 \neq 0$, $a = c_3/c_2 \neq 0$ and $\mathrm{Tr}_{q/2}(c_2^3/c_3^2) = 1$; then n ranges over q^3 elements, each with two roots. Contribution $2q^3 I(c_2, c_3)$.

Case IV: $a = 0$. $\alpha \in \mathbb{F}_{q^4}$, $(e_1, e_2, e_3) = (0, (\mathrm{Tr}\, \alpha)^2, 0)$; contribution $q^3[c_1 = 0][c_3 = 0]$.

Adding the four contributions gives the theorem. □

Corollary 3.3. *For every $q = 2^k$:*

(a) $\varphi = \Psi - q^5$ *is supported on* $\mathcal{G}_0 = \{c_1 = 0\}$.
(b) $\Psi(0,0,0) = \#\{\alpha \in \mathbb{F}_{q^8} : e_1 = e_2 = e_3 = 0\} = q^5 - q^4 + q^3$*; equivalently there are exactly* $q^4(q-1)/8$ *irreducible octics with* $a_7 = a_6 = a_5 = 0$.
(c) *For* $c \neq 0$, $\Psi(0,c,0) = q^5 + \varepsilon_0\, q^3$, *i.e.* $q^5 + q^3$ *for* k *even and* $q^5 - q^3$ *for* k *odd.*
(d) $A := \sum_c \Psi(0,c,0) = \#\{\alpha : \operatorname{Tr}\alpha = \operatorname{Tr}\alpha^3 = 0\}$ *equals* q^6 *for* k *even and* $q^6 - 2q^4 + 2q^3$ *for* k *odd.*

Proof. (a) is the first line of the theorem. (b): $K(0,0) = q^2/2$, $I(0,0) = 0$. (c): with $y = \sqrt{c}\,v$ one has $\chi(cv^2/w + \sqrt{c}\,v/w) = \chi((y^2+y)/w)$; since $y \mapsto y^2 + y$ is two-to-one onto $\{\operatorname{Tr} z = 0\}$ and $z \mapsto \operatorname{Tr}(z/w)$ is nontrivial there unless $w = 1$, $K(c,0) = q[1 \in W] = q[k \text{ odd}]$. (d) follows from (b),(c). The prime-power part of $\Psi(0,c,0)$ equals q^3 for every c (Part I), which gives the count of irreducibles in (b). □

Theorem 3.4. *For every* $q = 2^k$, $\dfrac{\operatorname{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1)$.

Proof. By Corollary 3.3(b),(c), $\sum_{g\in\mathcal{G}^2} \varphi(g)^2 = (q^4-q^3)^2 + (q-1)q^6 = q^7(q-1)$, and $|\mathcal{G}^2| = q$. □

Remark 3.5. The factor $q(q-1) = (q-1)^2 + (q-1)$ splits as the contribution of the identity class (the interval around t^8, whose prime deficit is $q^4 - q^3$, relative size $1/q$) plus that of the other $q-1$ Legendre classes ($\pm q^3$ each). The sign in (c) depends on the parity of k but the variance does not.

3.3. **Cubic Artin–Schreier sums over $\mathbb{F}_{q^8}$.** Let $S(b,a) = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(ax^3 + bx)$.

Lemma 3.6. *For* $a \in \mathbb{F}_q^\times$ *and* $b \in \mathbb{F}_q$, $S(b,a) = S(0,a)$.

Proof. For any $c \in \mathbb{F}_{q^8}$ the substitution $x \mapsto x + c$ gives $S(b,a) = \psi_0(ac^3 + bc)\, S(b + T_a(c), a)$ with $T_a(c) = ac^2 + \sqrt{ac}$, since $\operatorname{Tr}(acx^2) = \operatorname{Tr}(\sqrt{ac}\,x)$. If $c \in \mathbb{F}_{q^4}$ then $\operatorname{Tr}_{q^8/2}(ac^3 + bc) = 0$. It remains to find $c \in \mathbb{F}_{q^4}$ with $T_a(c) = b$. The map T_a preserves $\mathbb{F}_{q^4}$ and is self-adjoint for $\langle u, v\rangle = \operatorname{Tr}_{q^4/2}(uv)$, so $\operatorname{im} T_a = (\ker T_a)^\perp$ with $\ker T_a = \{0\} \cup \{u \in \mathbb{F}_{q^4} : u^3 = a^{-1}\}$. For such $u \neq 0$, $\zeta = u^{q-1}$ is a cube root of unity. If $\zeta = 1$, $u \in \mathbb{F}_q$ and $\operatorname{Tr}_{q^4/2}(bu) = \operatorname{Tr}_{q/2}(4bu) = 0$. If $\zeta \neq 1$ and k is even, $\zeta^q = \zeta$ forces $u^{q^4} = \zeta^4 u \neq u$, impossible; if k is odd, $\zeta^q = \zeta^2$ gives $u^{q^2} = u$, so $u \in \mathbb{F}_{q^2}$ and $\operatorname{Tr}_{q^4/2}(bu) = \operatorname{Tr}_{q^2/2}(2bu) = 0$. Hence $b \perp \ker T_a$. □

Proposition 3.7 (Carlitz's evaluation, elementary proof)**.** *For* $a \in \mathbb{F}_q^\times$ *let* $\nu = a$. *Then*

$$S(0,a) = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(ax^3) = \begin{cases} -2q^4, & a \text{ is a cube in } \mathbb{F}_{q^8}, \\ +q^4, & \text{otherwise,} \end{cases}$$

and $a \in \mathbb{F}_q^\times$ *is a cube in* $\mathbb{F}_{q^8}$ *iff* k *is odd or* a *is a cube in* $\mathbb{F}_q$.

Proof. By Lemma 3.6, $S(0,a) = q^{-1}\sum_{b\in\mathbb{F}_q} S(b,a) = \sum_{\alpha:\operatorname{Tr}\alpha=0} \psi_0(a\alpha^3) = \widehat{\Psi}(0,a)$ in the notation of (3) below, whose value is computed in Theorem 3.11. The cube criterion: $(q^8-1)/(q-1) \equiv 8 \equiv 2 \pmod 3$ when $q \equiv 1 \pmod 3$, while $3 \mid (q^8-1)/(q-1)$ when $q \equiv 2 \pmod 3$. □

Remark 3.8. The classical route [4] uses the cubic Gauss sum $g_4 = \sum_{y\in\mathbb{F}_4^\times} \chi_3(y)(-1)^{\operatorname{Tr} y} = 2$ and the Hasse–Davenport relation [13] $g_{4^m} = (-1)^{m-1}2^m$, giving $\sum_{x\in\mathbb{F}_{2^{2m}}} \psi_0(ax^3) = (-1)^{m+1}2^{m+1}$ for cubes and $(-1)^m 2^m$ otherwise; with $m = 4k$ this agrees with Proposition 3.7.

3.4. **The full-group variance.** Identify $\mathcal{G}_0 = \{(0,x,y)\}$ with $(\mathbb{F}_q^2, +)$ and set

$$\widehat{\Psi}(\mu,\nu) = \sum_{x,y\in\mathbb{F}_q} \Psi(0,x,y)\chi(\mu x + \nu y). \tag{3}$$

For $j \in \mathbb{F}_q^\times$ let $C_j(x) = x^3 + (1+j)x^2 + x + 1$, $R_j = \#\{x \in \mathbb{F}_q : C_j(x) = 0\}$, $L_j = \#\{x : C_j(x) = 0,\ \operatorname{Tr}_{q/2} x = 1\}$ and $L_j' = R_j - L_j$.

Lemma 3.9. *For $(\mu,\nu)\neq(0,0)$:*

$$\widehat{\Psi}(\mu,0)=-q^4\quad(\mu\neq0),\qquad \widehat{\Psi}(0,\nu)=q^4\big(1-R_0(\nu)-2[k\ \text{odd}]\big)\quad(\nu\neq0),$$
$$\widehat{\Psi}(\mu,\nu)=q^4\big(1-3L_j-L_j'\big)\qquad(\mu\nu\neq0,\ j=\mu^3/\nu^2),$$

where $R_0(\nu)=\#\{t:t^3=\nu\}$ equals 1 for k odd and $3[\nu$ is a cube$]$ for k even. In particular $\widehat{\Psi}(0,\nu)=-2q^4$ for k odd, and for k even $-2q^4$ or $+q^4$ according as ν is a cube in $\mathbb{F}_q$ or not.

Proof. Transform the three terms of Theorem 3.2. The term $q^3[y=0]$ gives $q^4[\mu=0]$. For K, the identity $\sum_x\chi(x\alpha+\sqrt{x}\,\beta)=q[\sqrt{\alpha}=\beta]$ yields $\widehat{K}(\mu,\nu)=q^2\#\{(v,w)\in\mathbb{F}_q\times W:\ v^3=\nu w^2,\ v^2(w+1)=\mu w^2\}$. For $I=[x,y\neq0]\frac{1-\chi(x^3/y^2)}{2}$, the substitution $x=yt$ gives $\sum_{x,y\neq0}\chi(x^3/y^2+\mu x+\nu y)=q\,R^*(\mu,\nu)-(q-1)$ with $R^*=\#\{t\neq0:t^3+\mu t+\nu=0\}$. The three cases follow; for $\mu\nu\neq0$, dividing the two conditions in $\widehat{K}$ gives $v=\nu(w+1)/\mu$ and $\nu^2(w+1)^3=\mu^3w^2$, and $t=(\nu/\mu)s$, $s=1+1/w$ transforms $t^3+\mu t+\nu=0$ into $C_j(w)=0$, so $R^*=R_j$ and $\widehat{K}=q^2L_j$. □

Lemma 3.10. $\Sigma:=\displaystyle\sum_{j\in\mathbb{F}_q^\times}(1-3L_j-L_j')^2=3q-4+(-1)^k$.

Proof. Put $c(x)=2-\chi(x)\in\{1,3\}$, so $3L_j+L_j'=\sum_{C_j(x)=0}c(x)$. Every $x\in\mathbb{F}_q\setminus\{0,1\}$ is a root of exactly one C_j ($j=(x+1)^3/x^2$), and $0,1$ are roots of none. Expanding, $\Sigma=(q-1)-2\sum_{x\neq0,1}c(x)+\sum_{x\neq0,1}c(x)^2+\mathrm{OFF}$, where OFF is the sum of $c(x)c(x')$ over ordered pairs of *distinct* roots of a common C_j. Using $\sum_{x\neq0,1}\chi(x)=-1-(-1)^k$ one finds $\Sigma=2q-1+2(-1)^k+\mathrm{OFF}$. Distinct roots $x\neq x'$ of a common C_j are exactly the pairs with $x,x'\notin\{0,1\}$ satisfying

$$x^2x'^2+xx'+x+x'=0 \tag{4}$$

(the third root is $1/(xx')$ and $e_2=1$; conversely (4) with $x,x'\notin\{0,1\}$ produces a cubic C_j with $j\neq0$). On (4) one has $x+x'=p^2+p$ with $p=xx'$, hence $\chi(x)\chi(x')=1$, and by symmetry $\mathrm{OFF}=5P-4\sum_{\mathrm{pts}}\chi(x)$ where P is the number of points. For fixed $x\notin\{0,1\}$, (4) is a quadratic in x' with $1+\chi(x^3/(x+1)^2)$ roots, all $\notin\{0,1\}$. With $y=x+1$: $x^3/(x+1)^2=y+1+1/y+1/y^2$, whose χ-value is $(-1)^k\chi(y)$, and $x+x^3/(x+1)^2=1/y+1/y^2$, whose χ-value is 1. Therefore $P=\sum_{\mathrm{pts}}\chi(x)=q-3-(-1)^k$, $\mathrm{OFF}=q-3-(-1)^k$ and $\Sigma=3q-4+(-1)^k$. □

Theorem 3.11. *For every $q=2^k$, $\dfrac{\mathrm{Var}_{\mathcal{G}}(\Psi)}{q^5}=3-\dfrac{3}{q}$ and $\dfrac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{\mathrm{Var}_{\mathcal{G}}(\Psi)}=\dfrac{q^2}{3}$.*

Proof. By Corollary 3.3(a) and Parseval on $\mathcal{G}_0\cong\mathbb{F}_q^2$ (note $\sum_{\mathcal{G}_0}\varphi=0$),

$$\sum_{\mathcal{G}}\varphi^2=\frac{1}{q^2}\sum_{(\mu,\nu)\neq(0,0)}|\widehat{\Psi}(\mu,\nu)|^2.$$

Lemma 3.9 gives $(q-1)q^8$ from $\nu=0$, $\kappa_k(q-1)q^8$ from $\mu=0$ with $\kappa_k=4$ (k odd) and $\kappa_k=\frac{1}{3}\cdot4+\frac{2}{3}\cdot1=2$ (k even), and $q^8(q-1)\Sigma$ from $\mu\nu\neq0$ (each j has exactly $q-1$ preimages (μ,ν)). By Lemma 3.10 the bracket $1+\kappa_k+\Sigma$ equals $3q$ in both parities, so $\sum_{\mathcal{G}}\varphi^2=3q^7(q-1)$. □

Theorem 3.12. *Let $\varepsilon_{\lambda_1,\lambda_3}$ be the quadratic characters of $\mathcal{G}_3$ (Proposition 2.2). Then*

(i) *$\rho(\varepsilon_{\lambda_1,0})=1$ for all $\lambda_1\neq0$ ($q-1$ characters of conductor u^2);*
(ii) *$\rho(\varepsilon_{\lambda_1,\lambda_3})=r(\lambda_3)$ is independent of λ_1 and constant on cosets of the cubes in $\mathbb{F}_q^\times$;*
(iii) *if k is odd, $r\equiv\dfrac{q-3}{3(q-1)}$;*
(iv) $\rho_{\mathrm{avg}}=\dfrac{q^2/3-1}{q^2-1}\to\dfrac{1}{3}$.

Proof. (i),(ii): φ is supported on $\mathcal{G}_0$ where $p_1=0$ and $p_3=c_3$, so $\varepsilon_{\lambda_1,\lambda_3}|_{\mathcal{G}_0}=\chi(\lambda_3c_3)$; the scaling automorphism $(c_1,c_2,c_3)\mapsto(sc_1,s^2c_2,s^3c_3)$ preserves Ψ. (iii): for k odd cubing is onto,

so $\Phi(y)=\sum_x \varphi(0,x,y)^2$ is constant for $y\neq 0$, equal to $(3q^7(q-1)-q^7(q-1))/(q-1)=2q^7$ by Theorems 3.4–3.11; then $r=(q^7(q-1)-2q^7)/(3q^7(q-1))$. (iv) is immediate. □

Remark 3.13. For $q=8$ this gives seven characters with $\rho=1$ and 56 with $\rho=5/21=0.2381$, matching [5, Table 4] exactly; the "order-two coordinates" there contribute $7\cdot 5/21=5/3$. The excess over Keating–Rudnick comes from the primitive quadratic characters, whose ρ does not decay in q; $\sum_{\chi\neq 1}\psi_8(\chi)=-q^7+q^6$ shows that $\mathbb{E}_\chi \operatorname{Tr}\Theta_\chi^8\to -1$, so the Frobenius classes are not Haar-distributed in $U(2)$ when $p=2$, $\ell=3$.

4. The case $d=6$: the three-value structure theorem

Now $n=12$, $\ell=5$, $\mathcal{G}^2=\{(0,c_2,0,c_4,0)\}$ has order q^2 and

$$U:=\{\alpha\in\mathbb{F}_{q^{12}}:\ \operatorname{Tr}\alpha=\operatorname{Tr}\alpha^3=\operatorname{Tr}\alpha^5=0\},\qquad \operatorname{Tr}=\operatorname{Tr}_{q^{12}/q}.$$

Since $p_3=c_3+c_1c_2+c_1^3$ and $p_5\equiv c_5$ modulo terms involving c_1,c_3, the Legendre classes are exactly the classes with $p_1=p_3=p_5=0$, and $\Psi(0,c_2,0,c_4,0)=\#\{\alpha\in U: e_2(\alpha)=c_2,\ e_4(\alpha)=c_4\}$. Define

$$A_6=|U|,\qquad X=\sum_{\alpha\in U}\chi(e_2(\alpha)),\qquad Y=\sum_{\alpha\in U}\chi(e_4(\alpha)).$$

Lemma 4.1. *U is stable under $\alpha\mapsto\alpha+c$ ($c\in\mathbb{F}_q$) and $\alpha\mapsto s\alpha$ ($s\in\mathbb{F}_q^\times$). On U,*

$$e_2(\alpha+c)=e_2(\alpha),\qquad e_4(\alpha+c)=e_4(\alpha)+c^2e_2(\alpha)+c^4,\qquad (e_2,e_4)(s\alpha)=(s^2e_2,s^4e_4)(\alpha).$$

Proof. $\operatorname{Tr}(\alpha+c)^3=\operatorname{Tr}\alpha^3+c(\operatorname{Tr}\alpha)^2$ and $\operatorname{Tr}(\alpha+c)^5=\operatorname{Tr}\alpha^5+c(\operatorname{Tr}\alpha)^4$ since $12c=0$. For the elementary symmetric functions, $e_j(\alpha+c)=\sum_i\binom{12-i}{j-i}c^{j-i}e_i(\alpha)$; modulo 2, $\binom{12}{2}=66$, $\binom{11}{1}=11$, $\binom{12}{4}=495$, $\binom{11}{3}=165$, $\binom{10}{2}=45$, $\binom{9}{1}=9$, and on U one has $e_1=e_3=0$. □

Theorem 4.2 (three-value profile)**.** *For all $q=2^k$ and all $(c_2,c_4)\in\mathbb{F}_q^2$,*

$$\Psi(0,c_2,0,c_4,0)=q^7+\frac{A_6-q^9-X}{q^2}+\frac{X}{q}\,[c_2=0]+\frac{Y}{q}\,[c_2\neq 0]\,\chi\Big(\frac{c_4}{c_2^2}\Big).$$

Equivalently, the Fourier transform $\widehat{\Psi}_H(\mu,\nu)=\sum_{\alpha\in U}\chi(\mu e_2+\nu e_4)$ equals A_6, X, $Y\chi(\mu/\sqrt{\nu})$ for $(\mu,\nu)=(0,0)$, $(\mu\neq 0,0)$, $\nu\neq 0$ respectively.

Proof. Let $F(\lambda)=\sum_U\chi(\lambda e_2+e_4)$. By Lemma 4.1, $F(\lambda)=\sum_U\chi(\lambda e_2+e_4+c^2e_2+c^4)=\chi(c)F(\lambda+c^2)$, hence $F(\lambda)=\chi(\sqrt{\lambda})F(0)=\chi(\lambda)Y$. For $\nu\neq 0$ choose $s=\nu^{-1/4}$; scaling gives $\widehat{\Psi}_H(\mu,\nu)=F(\mu\nu^{-1/2})=\chi(\mu/\sqrt{\nu})Y$, and for $\mu\neq 0$ scaling gives $\widehat{\Psi}_H(\mu,0)=X$. Fourier inversion on $\mathbb{F}_q^2$ yields the displayed formula. □

Corollary 4.3. *Put $B=(A_6-q^9-X)/q^2$. Then $\varphi=\Psi-q^7$ equals $\varphi_0=B+X/q$ on the q classes with $c_2=0$, $\varphi_A=B+Y/q$ on the $q(q-1)/2$ classes with $c_2\neq 0$, $\operatorname{Tr}_{q/2}(c_4/c_2^2)=0$, and $\varphi_B=B-Y/q$ on the remaining $q(q-1)/2$ classes, and*

$$\frac{\operatorname{Var}_{\mathcal{G}^2}(\Psi)}{q^7}=\frac{q\varphi_0^2+q(q-1)\big(B^2+Y^2/q^2\big)}{q^9}. \tag{5}$$

Remark 4.4. The same argument applies to every d: $U_d=\{p_j=0,\ j\text{ odd}\le\ell\}$ is stable under $\mathbb{F}_q$-translations and scalings, and the transformation law of e_{2j} is governed by $\binom{2d-i}{2j-i}\bmod 2$ (Lucas' theorem). For $d=4$ one recovers Corollary 3.3(c).

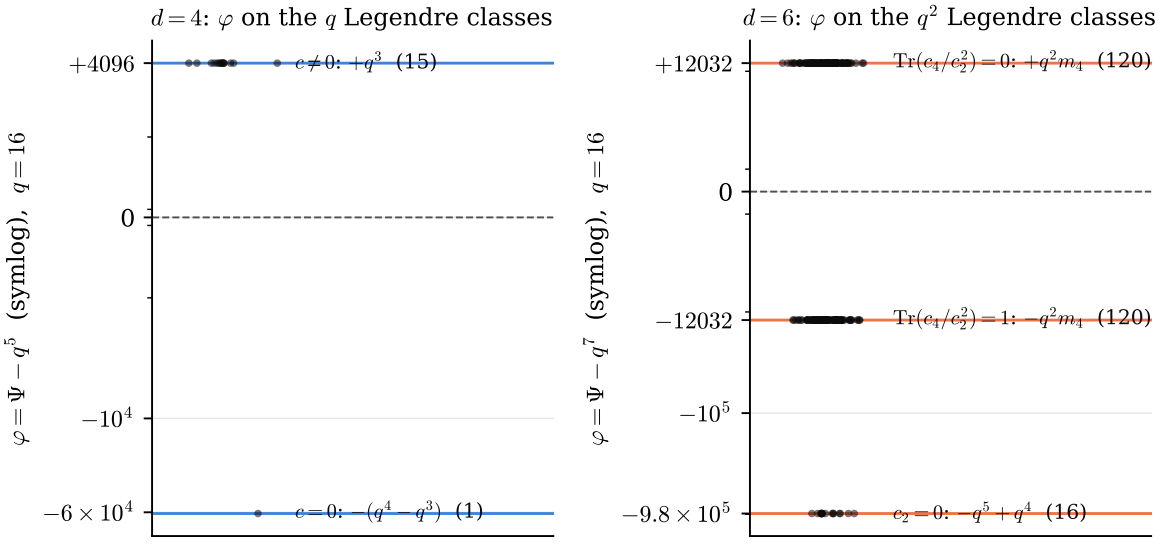


FIGURE 1. The Legendre deviations $\varphi=\Psi-q^{d+1}$ for $q=16$. Left: $d=4$, the q classes $(0,c,0)$ take the two values of Corollary 3.3; the identity class carries $-(q^4-q^3)$. Right: $d=6$, the q^2 classes $(0,c_2,0,c_4,0)$ take exactly the three values of Theorem 4.2 ($m_4=47$). Each dot is one class (horizontal jitter only); the deviations are not continuous noise but are pinned to discrete levels by the translation symmetry of U.

5. The quadratic tower $\mathbb{F}_{q^{12}}\supset\mathbb{F}_{q^6}$

Lemma 5.1. *For $\alpha\in\mathbb{F}_{q^{12}}$ let $a=\alpha+\alpha^{q^6}$, $n=\alpha^{1+q^6}\in\mathbb{F}_{q^6}$, $\mathrm{Tr}=\mathrm{Tr}_{q^6/q}$, and $e_j(a)$ the elementary symmetric functions of the six conjugates of a. Then*

$$\begin{aligned}
e_1&=\mathrm{Tr}\,a,\qquad e_2=e_2(a)+\mathrm{Tr}\,n,\qquad e_3=e_3(a)+\mathrm{Tr}\,a\,\mathrm{Tr}\,n+\mathrm{Tr}(an),\\
e_4&=e_4(a)+e_2(a)\,\mathrm{Tr}\,n+e_1(a)\,\mathrm{Tr}(an)+\mathrm{Tr}(a^2n)+e_2(n),\\
e_5&=e_5(a)+e_3(a)\,\mathrm{Tr}\,n+e_2(a)\,\mathrm{Tr}(an)+e_1(a)\,\mathrm{Tr}(a^2n)+\mathrm{Tr}(a^3n)\\
&\qquad+e_1(a)e_2(n)+\mathrm{Tr}(an)\,\mathrm{Tr}\,n+\mathrm{Tr}(an^2),\\
p_3&=\mathrm{Tr}(a^3+an),\qquad p_5=\mathrm{Tr}(a^5+a^3n+an^2),
\end{aligned}$$

and $e_2(\beta+\beta^q)=\mathrm{Tr}_{q^{12}/q}(\beta^{q+1})+(\mathrm{Tr}_{q^{12}/q}\,\beta)^2$ for all $\beta\in\mathbb{F}_{q^{12}}$. The multiplicity statement of Lemma 3.1 holds verbatim with $\mathbb{F}_{q^4}$ replaced by $\mathbb{F}_{q^6}$; in particular $\mathbb{F}_{q^6}\subset U$, with $e_2|_{\mathbb{F}_{q^6}}=(\mathrm{Tr}\,\alpha)^2$ and $e_4|_{\mathbb{F}_{q^6}}=e_2(\alpha)^2$.

Proof. Expand $\prod_{i<6}(t^2+a_it+n_i)$ using $e_2(a\backslash j)=e_2(a)+a_je_1(a)+a_j^2$, $e_3(a\backslash j)=e_3(a)+a_je_2(a)+a_j^2e_1(a)+a_j^3$ and $e_1(a\setminus j,k)=e_1(a)+a_j+a_k$. The power sums follow from $s_k=as_{k-1}+ns_{k-2}$ for $s_k=\alpha^k+\alpha^{q^6k}$. All identities were also checked by exhaustive computation over $\mathbb{F}_{2^{12}}$. □

Lemma 5.2 (Gauss sum of e_2)**.** *On $\mathbb{F}_{q^6}$ the quadratic form e_2 has polar form $B(x,y)=\mathrm{Tr}\,x\,\mathrm{Tr}\,y+\mathrm{Tr}(xy)$, which is nondegenerate. For every $\beta\in\mathbb{F}_{q^6}$, $\sum_{n\in\mathbb{F}_{q^6}}\chi(e_2(n)+\mathrm{Tr}(\beta n))=\chi(e_2(\beta))\,G$ with $G=\sum_n\chi(e_2(n))$.*

Proof. $n_0=\beta+\mathrm{Tr}\,\beta$ satisfies $B(n_0,\cdot)=\mathrm{Tr}(\beta\,\cdot)$, and $e_2(\beta+c)=e_2(\beta)+c\,\mathrm{Tr}\,\beta+c^2$ for $c\in\mathbb{F}_q$ gives $e_2(n_0)=e_2(\beta)$. □

Theorem 5.3 (tower reduction)**.** *Let a range over $\mathbb{F}_{q^6}\backslash\{0\}$ with $\mathrm{Tr}\,a=0$, $(\lambda,\mu)\in\mathbb{F}_q^2$, $\varepsilon\in\{0,1\}$, and put*

$$\beta'=\lambda a+\sqrt{\mu}\sqrt{a}+\mu a^3+\varepsilon a^{-2},\qquad \beta=e_2(a)+a^2+\beta',\qquad w=\chi(\lambda\,\mathrm{Tr}\,a^3+\mu\,\mathrm{Tr}\,a^5).$$

Then

$$A_6 = q^6 + q^4 \sum_a \sum_{\beta'=0} (-1)^{\varepsilon} w, \qquad X = q^4 \sum_a \chi(e_2(a)) \sum_{\beta'=1} (-1)^{\varepsilon} w,$$
$$Y = G\Big[1 + q^{-2} \sum_a \sum_{\lambda,\mu,\varepsilon} (-1)^{\varepsilon} \chi\big(e_4(a) + \lambda \operatorname{Tr} a^3 + \mu \operatorname{Tr} a^5 + e_2(\beta)\big)\Big].$$

Proof. For $a \neq 0$ with $\operatorname{Tr} a = 0$, the conditions $p_3 = p_5 = 0$ read $\operatorname{Tr}(an) = \operatorname{Tr} a^3$ and $\operatorname{Tr}(\sqrt{a}\, n)^2 + \operatorname{Tr}(a^3 n) = \operatorname{Tr} a^5$ (using $\operatorname{Tr}(an^2) = \operatorname{Tr}(\sqrt{a}\, n)^2$); expand their indicators and the irreducibility indicator $\frac{1}{2}\sum_{\varepsilon}(-1)^{\varepsilon}\chi(\varepsilon \operatorname{Tr}(a^{-2}n))$ by additive characters, using $\chi(\mu t^2) = \chi(\sqrt{\mu}\, t)$. The n-sums are $q^6[\gamma = 0]$ for A_6, X (no $e_2(n)$ term) and $\chi(e_2(\beta))G$ for Y by Lemma 5.2. The contribution of $\alpha \in \mathbb{F}_{q^6}$ is q^6, 0, G respectively. □

Remark 5.4. The equation $\beta' = 0$ with $\varepsilon = 0$ and $\mu \neq 0$ squares to $\mu^2 a^5 + \lambda^2 a + \mu = 0$, forcing $a \in \mathbb{F}_{q^2} \cup \mathbb{F}_{q^3}$ (a thin layer, as in Case II of Theorem 3.2). For $\varepsilon = 1$, with $u = \mu a^5 + \lambda a^3 + 1$ it becomes $u^2 + u = \lambda a^3 + 1$: the points $(a, u) \in E_\lambda(\mathbb{F}_{q^6})$ of the supersingular elliptic curve $E_\lambda : y^2 + y = \lambda x^3 + 1$ with $(u + \lambda a^3 + 1)/a^5 \in \mathbb{F}_q$. The additional term $e_2(\beta)$ in Y is quadratic in $(\lambda, \sqrt{\mu})$ with cubic cross terms, which is why Y does not collapse onto a thin layer.

The reductions were verified numerically for $q = 2, 4, 8$; they give $G = \varepsilon_0 q^3$ for $k \leq 3$.

6. Quadratic forms and supersingular Artin–Schreier curves

6.1. A_6 and X as sums of Frobenius traces.

Theorem 6.1. *Write $\lambda = (\lambda_1, \lambda_3, \lambda_5) \in \mathbb{F}_q^3$ and $S(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$. Then*

$$A_6 - q^9 = \frac{1}{q^3} \sum_{\lambda \neq 0} S(\lambda), \tag{6}$$

and $S(\lambda) = -\sum_j \omega_j(\lambda)^{12}$, where $\omega_j(\lambda)$ are the Frobenius eigenvalues over $\mathbb{F}_q$ of the Artin–Schreier curve $C_\lambda : y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ (genus 2, 1, 0 according as $\lambda_5 \neq 0$; $\lambda_5 = 0 \neq \lambda_3$; only $\lambda_1 \neq 0$). Moreover $\operatorname{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3) = Q_\lambda(x)$ is an $\mathbb{F}_2$-quadratic form with polar form $B_\lambda(x, y) = \operatorname{Tr}(y\, T_\lambda(x))$,

$$T_\lambda(x) = \lambda_5 x^4 + (\lambda_5 x)^{1/4} + \lambda_3 x^2 + (\lambda_3 x)^{1/2}, \qquad R_\lambda = \ker T_\lambda = \{x : \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_3^2 x^2 + \lambda_5 x = 0\},$$

so that $S(\lambda) \in \{0, \pm 2^{(12k + r_\lambda)/2}\}$ with $r_\lambda = \dim_{\mathbb{F}_2} R_\lambda \leq 4$ (for the Gold and Kasami–Welch monomials this is the setting of [16]), $S(\lambda) = 0$ iff $\operatorname{Tr}(\lambda_1 x) \neq Q_\lambda(x)$ for some $x \in R_\lambda$, and the sign otherwise is the Arf invariant of Q_λ on $\mathbb{F}_{q^{12}}/R_\lambda$ (shifted by $Q_\lambda(x_0)$, $B_\lambda(x_0, \cdot) = \operatorname{Tr}(\lambda_1 \cdot)$). In particular all C_λ are supersingular: $\omega_j(\lambda) = \sqrt{q}\, \zeta_j$ with roots of unity ζ_j.

Proof. $A_6 = \sum_{g \in \mathcal{G}^2} \Psi(g) = \frac{|\mathcal{G}^2|}{|\mathcal{G}|} \sum_{\varepsilon^2 = 1} \psi_{12}(\varepsilon)$ and $\psi_{12}(\varepsilon_\lambda) = S(\lambda)$ by Proposition 2.2; $S(0) = q^{12}$. The L-function interpretation is standard. $Q_\lambda(x) = \operatorname{Tr}(x(\lambda_5 x^4 + \lambda_3 x^2))$ is of the form $\operatorname{Tr}(xL(x))$ with L linearised, whence the quadratic-form statements; the kernel of T_λ equals that of T_λ^4. Since the same structure persists over every $\mathbb{F}_{q^m}$, $\sum_{x \in \mathbb{F}_{q^m}} (-1)^{\operatorname{Tr} f_\lambda(x)} \in \{0, \pm 2^{(mk+r)/2}\}$ for all m, which forces the normalised eigenvalues to be roots of unity [11]. □

Summing over λ_1 and using Hilbert 90 ($\alpha = \beta + \beta^q$, $\theta = \sigma + \sigma^{-1}$, $\operatorname{Tr} \alpha^3 = \operatorname{Tr}(\beta^2 \theta \beta)$, $\operatorname{Tr} \alpha^5 = \operatorname{Tr}(\beta^4 \theta \beta)$) one obtains the equivalent forms

$$A_6 = q^9 + \frac{1}{q^3} \sum_{(\lambda_3, \lambda_5) \neq 0} W(\lambda_3, \lambda_5), \qquad X = \frac{1}{q^3} \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5), \tag{7}$$

with $W = \sum_\beta (-1)^{\operatorname{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta)}$ and $W' = \sum_\beta (-1)^{\operatorname{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta + \beta^{q+1} + \beta)}$ (the second uses Lemma 5.1). Their polar forms are $\operatorname{Tr}(\gamma M(\theta\beta))$ with $M(y) = \lambda_3 y^2 + \sqrt{\lambda_3 y} + \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ ($+y$ for W'), so the radicals are $\theta^{-1}(\ker M \cap \operatorname{im} \theta)$, of dimension $2k + \dim(\ker M \cap \operatorname{im} \theta)$, where $\ker \theta = \mathbb{F}_{q^2}$ and $\operatorname{im} \theta = \ker \operatorname{Tr}_{q^{12}/q^2}$.

6.2. **The pure cubic part.**

Lemma 6.2. *For all k and all $\lambda_3 \in \mathbb{F}_q^\times$: $W(\lambda_3, 0) = 0$ if k is odd and $W(\lambda_3, 0) = -2q^7$ if k is even.*

Proof. $\ker M = \{0\} \cup \{y : y^3 = \lambda_3^{-1}\}$, and all three cube roots lie in $\mathbb{F}_{q^{12}}$. With $\zeta = y^{q-1} \in \mu_3$: $\zeta = 1$ gives $y \in \mathbb{F}_q$; for $\zeta \neq 1$, k odd gives $y \in \mathbb{F}_{q^2}$ and k even gives $y \in \mathbb{F}_{q^3}$. In all cases $\mathrm{Tr}_{q^{12}/q^2} y = 0$, so $\ker M \subset \mathrm{im}\,\theta$ and $\dim R = 2k + 2$, $|W| \in \{0, 2^{7k+1}\}$. Now $Q(\beta) = \mathrm{Tr}_{q^{12}/2}(\lambda_3 \beta^2 \theta\beta)$ vanishes on $\ker\theta$, and for $\theta\beta = y$ it is independent of the choice of β (the ambiguity $c \in \mathbb{F}_{q^2}$ contributes $\mathrm{Tr}(\lambda_3 c^2 y)$ with $c^2 y$ in a subfield of even index). If $y \in \mathbb{F}_{q^2}$, $\zeta \neq 1$: on $\mathbb{F}_{q^4}$ one has $\theta\beta = \sigma(\mathrm{Tr}_{q^4/q^2}\,\beta)$, so pick $\beta_y \in \mathbb{F}_{q^4}$ with $\mathrm{Tr}_{q^4/q^2}\,\beta_y = y^q$; then $Q(\beta_y) = \mathrm{Tr}_{q^4/2}(\lambda_3 y \beta_y^2) = \mathrm{Tr}_{q^2/2}(\lambda_3 y\,(\mathrm{Tr}_{q^4/q^2}\,\beta_y)^2) = \mathrm{Tr}_{q^2/2}(\lambda_3 y^{2q+1}) = \mathrm{Tr}_{q^2/2}(\zeta^2) = k \pmod 2$, which is 1 for k odd: $Q|_R \neq 0$ and $W = 0$. If $y \in \mathbb{F}_q$ the same computation gives $\mathrm{Tr}_{q^2/2}(1) = 0$; if $y \in \mathbb{F}_{q^3}$ (k even) then $\mathrm{Tr}_{q^3/q}\, y = 0$, θ is invertible on the trace-zero part of $\mathbb{F}_{q^3}$, $\beta_y \in \mathbb{F}_{q^3}$ and $Q(\beta_y) = 0$. So for k even $|W| = 2q^7$. Finally $W = \sum_{\lambda_1 \in \mathbb{F}_q} S(\lambda_1, \lambda_3, 0)$ with $|S| = 2q^6$ for each term ($r = 2$), and $|W| = q \cdot 2q^6$ forces all q signs to agree with that of $S(0, \lambda_3, 0) = -2q^6$ (Hasse–Davenport, $m = 6k$). □

Remark 6.3. For k odd the analogue of Lemma 3.6 fails over $\mathbb{F}_{q^{12}}$: the cube roots lie in $\mathbb{F}_{q^2}$ and $[\mathbb{F}_{q^6} : \mathbb{F}_{q^2}] = 3$ is odd, so $\mathbb{F}_q \not\subset T(\mathbb{F}_{q^6})$. The vanishing of $W(\lambda_3, 0)$ is exactly this failure.

6.3. **The pure quintic part.** The quintic analogue uses a pure Gauss sum of order 5. Since $2^2 \equiv -1 \pmod 5$, Stickelberger's theorem on pure Gauss sums (see [2, Thm. 11.6.3]) gives, for every character χ of order 5 of $\mathbb{F}_{2^{4s}}^\times$, $G(\chi) = \sum_y \chi(y)(-1)^{\mathrm{Tr}\, y} = (-1)^{s-1} 2^{2s}$. With $4s = 12k$, i.e. $s = 3k$, and $a \in \mathbb{F}_q^\times$,

$$S(0, 0, a) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(ax^5) = \sum_{j=1}^{4} \chi^{-j}(a) G(\chi^j) = \varepsilon_0\, q^6 \big(5\,[a \in (\mathbb{F}_{q^{12}}^\times)^5] - 1\big), \tag{8}$$

and $a \in \mathbb{F}_q^\times$ is a fifth power in $\mathbb{F}_{q^{12}}$ iff $4 \nmid k$ or $a \in (\mathbb{F}_q^\times)^5$ (because $(q^{12} - 1)/(q - 1) \equiv 0 \pmod 5$ iff $q \not\equiv 1 \pmod 5$ iff $4 \nmid k$).

Lemma 6.4 (pure quintic part)**.** *For all k and all $\lambda_5 \in \mathbb{F}_q^\times$:*

$$W(0, \lambda_5) = \begin{cases} 2q^7, & k \text{ odd}, \\ 0, & k \equiv 2 \pmod 4, \\ -4q^7, & k \equiv 0 \pmod 4 \text{ and } \lambda_5 \in (\mathbb{F}_q^\times)^5, \\ +q^7, & k \equiv 0 \pmod 4 \text{ and } \lambda_5 \notin (\mathbb{F}_q^\times)^5. \end{cases}$$

Proof. Here $M(y) = \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ and $\ker M = \{0\} \cup \{y : y^{15} = \lambda_5^{-3}\}$. The equation $y^{15} = \lambda_5^{-3}$ is solvable in $\mathbb{F}_{q^{12}}$ iff λ_5 is a fifth power there (if $y^{15} = w^3$ then $y^5 = w\omega$ with $\omega^3 = 1$, and $\omega \in (\mathbb{F}_{q^{12}}^\times)^5$), and then, with $y_0^5 = \lambda_5^{-1}$, $\ker M = y_0 \mathbb{F}_{16}$ since $\mu_{15} \cup \{0\} = \mathbb{F}_{16} \subset \mathbb{F}_{q^{12}}$.

Case $4 \mid k$, $\lambda_5 \notin (\mathbb{F}_q^\times)^5$. $\ker M = 0$, the radical of the β-form is $\ker\theta = \mathbb{F}_{q^2}$ of dimension $2k$, Q vanishes on it, so $|W| = 2^{(12k+2k)/2} = q^7$. Each $S(\lambda_1, 0, \lambda_5)$ has trivial radical, hence $|S| = q^6$ for all q values of λ_1; $|W| = q \cdot q^6$ forces all signs to agree with that of $S(0, 0, \lambda_5) = -\varepsilon_0 q^6 = +q^6$ by (8). So $W = q^7$.

Case $\lambda_5 \in (\mathbb{F}_{q^{12}}^\times)^5$, k even. If $4 \mid k$ choose $y_0 \in \mathbb{F}_q$; if $k \equiv 2 \pmod 4$ then $5 \mid q + 1$, so every element of $\mathbb{F}_q$ is a fifth power in $\mathbb{F}_{q^2}$ and we choose $y_0 \in \mathbb{F}_{q^2}$. Also $\mathbb{F}_{16} \subset \mathbb{F}_{q^2}$. Hence $\ker M = y_0 \mathbb{F}_{16} \subset \mathbb{F}_{q^2} \subset \mathrm{im}\,\theta$ and the radical $R = \theta^{-1}(y_0 \mathbb{F}_{16})$ has dimension $2k+4$. For $y \in y_0 \mathbb{F}_{16}$ pick $\beta_y \in \mathbb{F}_{q^4}$ with $\mathrm{Tr}_{q^4/q^2}\,\beta_y = y^q$ (as in Lemma 6.2); then $Q(\beta_y) = \mathrm{Tr}_{q^2/2}(\lambda_5 y^{4q+1})$. Write $y = y_0 c$, $c \in \mathbb{F}_{16}$, and $\rho = y_0^{q-1} \in \mu_5$ ($\rho = 1$ if $4 \mid k$). Then $\lambda_5 y^{4q+1} = \rho^4 c^{4q+1}$, and $c^{4q+1} = c^5$ if $q \equiv 1 \pmod{15}$ ($4 \mid k$), $= c^2$ if $q \equiv 4 \pmod{15}$ ($k \equiv 2 \pmod 4$). Since $c \in \mathbb{F}_{16} \subset \mathbb{F}_{q^2}$, $\mathrm{Tr}_{q^2/2}(x) = \frac{k}{2}\,\mathrm{Tr}_{16/2}(x)$ for $x \in \mathbb{F}_{16}$. If $4 \mid k$ this is 0: $Q|_R = 0$, $|W| = 2^{(14k+4)/2} = 4q^7$; on the x-side the radical $y_0 \mathbb{F}_{16} \subset \mathbb{F}_q$ carries $Q_x(y_0 c) = \mathrm{Tr}_{q^{12}/2}(c^5) = 3k\,\mathrm{Tr}_{16/2}(c^5) = 0$ ($c^5 \in \mathbb{F}_4$)

and the linear forms $\mathrm{Tr}(\lambda_1 y_0 c) = \mathrm{Tr}_{q/2}(12\lambda_1 y_0 c) = 0$, so all q sums $S(\lambda_1, 0, \lambda_5)$ have modulus $2^{(12k+4)/2} = 4q^6$ and $|W| = q \cdot 4q^6$ forces equal signs, that of $S(0,0,\lambda_5) = 4\varepsilon_0 q^6 = -4q^6$: $W = -4q^7$. If $k \equiv 2 \pmod 4$ then $Q(\beta_{y_0 c}) = \mathrm{Tr}_{16/2}(\rho^4 c^2) = \mathrm{Tr}_{16/2}(\rho^2 c)$, which is nonzero for suitable c; so $Q|_R \neq 0$ and $W = 0$.

Case k odd. Fifth powers are bijective on $\mathbb{F}_q^\times$, so $y_0 \in \mathbb{F}_q$. Now $\mathbb{F}_{16} \not\subset \mathbb{F}_{q^2}$ and $q^2 \equiv 4 \pmod{15}$, so for $c \in \mathbb{F}_{16}$, $\mathrm{Tr}_{q^{12}/q^2}(y_0 c) = y_0 \sum_{i<6} c^{4^i} = y_0(c + c^4)$, which vanishes iff $c \in \mathbb{F}_4$. Thus $\ker M \cap \mathrm{im}\,\theta = y_0 \mathbb{F}_4$ and $\dim R = 2k+2$. For $y = y_0 c$, $c \in \mathbb{F}_4^\times$: $Q(\beta_y) = \mathrm{Tr}_{q^2/2}(\lambda_5 y_0^5 c^{4q+1}) = \mathrm{Tr}_{q^2/2}(1) = 0$ because $4q+1 \equiv q+1 \equiv 0 \pmod 3$. Hence $|W| = 2^{(14k+2)/2} = 2q^7$. On the x-side, $Q_x(y_0 c) = \mathrm{Tr}_{q^{12}/2}(c^5) = 0$ as before, while $\mathrm{Tr}_{q^{12}/2}(\lambda_1 y_0 c) = \mathrm{Tr}_{q/2}(\lambda_1 y_0)\,\mathrm{Tr}_{16/2}(c)$ (the twelve conjugates of c under q-Frobenius run three times through c, c^2, c^4, c^8); so exactly the $q/2$ values of λ_1 with $\mathrm{Tr}_{q/2}(\lambda_1 y_0) = 0$ give $|S| = 4q^6$, the others give 0, and $|W| = \frac{q}{2} \cdot 4q^6$ forces equal signs, that of $S(0,0,\lambda_5) = 4\varepsilon_0 q^6 = +4q^6$. So $W = 2q^7$. □

Lemmas 6.2 and 6.4 give, for the pure terms, $\sum_{\lambda_3 \neq 0} W(\lambda_3, 0) + \sum_{\lambda_5 \neq 0} W(0, \lambda_5) = 2(q-1)q^7$ for k odd and $= -2(q-1)q^7$ for k even (for $4 \mid k$ the quintic terms contribute $\frac{q-1}{5}(-4q^7) + \frac{4(q-1)}{5} q^7 = 0$). Hence Conjecture 6.5 for A_6 is equivalent to

$$\sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = \begin{cases} (q-1)(q-2)\, q^7, & k \text{ odd}, \\ -(q-1)(q-2)\, q^7, & k \text{ even}. \end{cases} \tag{9}$$

6.4. **Exact evaluation and the conjecture.** Every W, W' is the exponential sum of an $\mathbb{F}_2$-quadratic form (plus a linear form) on $\mathbb{F}_2^{12k}$, hence computable exactly by linear algebra: radical, test of $Q + L$ on it, a symplectic basis of a complement and its Arf invariant. We implemented this (`legendre_d6_qf.py`) with $\mathbb{F}_{q^{12}} = \mathbb{F}_q[z]/(m)$ and the maps $\beta \mapsto \beta^2, \beta^{1/2}, \sigma, \theta$ as $12k \times 12k$ matrices over $\mathbb{F}_2$.

Conjecture 6.5. *For all $q = 2^k$: $A_6 = q^9 + \varepsilon_0(q-1)q^5$ and $X = \varepsilon_0\, q^6 + q^5$, i.e.*

$$\sum_{\lambda_5 \neq 0} W(0, \lambda_5) + \sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = (q-1)q^7(\varepsilon_0 q + 2), \qquad \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5) = q^3(\varepsilon_0 q^6 + q^5).$$

The statement for A_6 with k odd is proved below (Corollary 6.13); the statement for X (analysed in §6.7), and that for A_6 with k even, remain conjectural.

TABLE 1. Exact values of A_6 and X by the quadratic-form algorithm ($N = 12k$ bits). For $k \leq 5$ the values coincide with the Hayes-group FFT of [5]; $k = 6, 7$ are beyond its reach at $d = 6$. In all cases $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$.

k	q	N	A_6	X	time
1	2	12	544	96	< 1 s
2	4	24	259 072	−3 072	< 1 s
3	8	36	134 447 104	294 912	< 1 s
4	16	48	68 703 748 096	−15 728 640	2 s
5	32	60	35 185 412 276 224	1 107 296 256	9 s
6	64	72	18 014 330 863 747 072	−67 645 734 912	48 s
7	128	84	9 223 376 400 541 548 544	4 432 406 249 472	225 s

Proposition 6.6 (structure of the data)**.** *For $k \leq 7$ the following hold.*

(a) *k odd: $W(0, \lambda_5) = 2q^7$ for all $\lambda_5 \neq 0$; the mixed W depends only on $j = \lambda_3^5/\lambda_5^3 \in \mathbb{F}_q^\times$ and equals $2q^7$ for exactly $(q-2)/2$ values of j and 0 otherwise. Hence $\sum W = (q-1)\,2q^7 + \frac{(q-1)(q-2)}{2}\,2q^7 = (q-1)q^8$.*

TABLE 2. Distribution of $W(\lambda_3,\lambda_5)/q^7$ over the q^2-1 nontrivial pairs (radical dimension in parentheses); "pure cubic" $\lambda_5=0$, "pure quintic" $\lambda_3=0$, "mixed" $\lambda_3\lambda_5\neq 0$.

k	pure cubic $(q-1)$	pure quintic $(q-1)$	mixed $((q-1)^2)$
1	0 (4)	2 (4)	0 (6) ×1
3	0 (8)	2 (8)	2 (8) ×21; 0 (10) ×28
5	0 (12)	2 (12)	2 (12) ×465; 0 (14) ×496
7	0 (16)	2 (16)	2 (16) ×8001; 0 (18) ×8128
2	−2 (6)	0 (8)	−4 (8) ×3; 1 (4) ×6
4	−2 (10)	−4 (12) ×3; 1 (8) ×12	−4 (12) ×75; 1 (8) ×90; 0 (12) ×60
6	−2 (14)	0 (16)	−4 (16) ×1386; 1 (12) ×1638; 0 (16) ×945

(b) *k even: $W(0,\lambda_5)=0$ if $k\equiv 2 \pmod 4$, while for $k\equiv 0 \pmod 4$ (so $5\mid q-1$) it equals $-4q^7$ for the $(q-1)/5$ fifth powers and $+q^7$ otherwise; the mixed W take values $-4q^7$, q^7, 0 with multiplicities $(q-1)(n_-,n_+,n_0)$, $(n_-,n_+,n_0)=(1,2,0),(5,6,4),(22,26,15)$ for $k=2,4,6$, satisfying $n_+-4n_-=2-q$.*

The set of j in (a) is not of the form $\{\mathrm{Tr}_{q/2}(j^m)=c\}$ for any m when $k=5,7$ (it is $\{\mathrm{Tr}\, j=0\}$ when $k=3$).

6.5. **The mixed part for odd k: a Gold-type quintic and a $\mathbb{Z}/4$ condition.** For k odd we describe the mixed terms completely in terms of the factorisation of a single quintic over $\mathbb{F}_q$; the evaluation is completed in Theorem 6.12 below.

Proposition 6.7. *Let k be odd, $\lambda_3\lambda_5\neq 0$, $j=\lambda_3^5/\lambda_5^3$, $t=j^{1/5}\in\mathbb{F}_q$ and $c=j^{-1/2}\in\mathbb{F}_q^\times$ (fifth and square roots are unique in $\mathbb{F}_q$).*

(a) *$W(\lambda_3,\lambda_5)$ depends only on j (the orbits of $(\lambda_3,\lambda_5)\mapsto(s^3\lambda_3,s^5\lambda_5)$, $s\in\mathbb{F}_q^\times$, are the fibres of j).*

(b) *With $y=\lambda_5^{-1/5}u$, the kernel equation $\lambda_5^4y^{15}+\lambda_3^4y^7+\lambda_3^2y+\lambda_5=0$ becomes $P_j(u):=u^{15}+t^4u^7+t^2u+1=0$, and*

$$P_j(u)=(u^5+t^2u+1)(u^{10}+t^2u^6+u^5+1),\qquad u=t^{1/2}v\implies u^5+t^2u+1=t^{5/2}(v^5+v+c).$$

The five roots $u_0,\dots,u_4$ of u^5+t^2u+1 have sum 0, are $\mathbb{F}_2$-linearly independent apart from this relation, and $\ker L_\lambda=\{0\}\cup\{u_i\}\cup\{u_i+u_{i'}\}$ (the ten roots of the second factor are the pairwise sums). The q-Frobenius acts on $\ker L_\lambda$ through its permutation action on the roots v_i of v^5+v+c, i.e. through the standard embedding $S_5\to GL_4(\mathbb{F}_2)$ on the sum-zero module.

(c) *Let $g\in S_5$ be the Frobenius permutation of the roots of v^5+v+c (its cycle type is the factorisation type of v^5+v+c over $\mathbb{F}_q$). Then $\ker L_\lambda\subset\mathbb{F}_{q^{12}}$ iff g has no 5-cycle, $\dim_{\mathbb{F}_2}(\ker L_\lambda\cap\operatorname{im}\theta)=2$ if g is a 4-cycle (type $(1,4)$) and $=4$ for types $(1,1,1,2)$, $(2,3)$, $(1,1,1,1,1)$, $(1,2,2)$, $(1,1,3)$, and the image of $\mathrm{Tr}_{q^{12}/q}$ on $\ker L_\lambda$ is one-dimensional over $\mathbb{F}_2$ if g is a 4-cycle and zero otherwise. Consequently, for a 4-cycle, $|W|\in\{0,2q^7\}$ and exactly $q/2$ of the sums $S(\lambda_1,\lambda_3,\lambda_5)$ are nonzero (with modulus $4q^6$); for the other types $|W|\in\{0,4q^7\}$ and either all or none of the $S(\lambda_1,\cdot)$ vanish.*

Proof. (a) For $s\in\mathbb{F}_q^\times$, $\beta\mapsto s\beta$ sends $\mathrm{Tr}(\lambda_3\beta^2\theta\beta+\lambda_5\beta^4\theta\beta)$ to the form with $(s^3\lambda_3,s^5\lambda_5)$; since $3,5\nmid q-1$ the map $s\mapsto(s^3,s^5)$ is injective, so the orbits have size $q-1$ and are the fibres of j. (b) Substitute $y=\lambda_5^{-1/5}u$ and divide by λ_5. The factorisation is checked by expanding: $u^5P_j(u)=(u^{10}+t^2u^6+u^5)^2+(u^{10}+t^2u^6+u^5)$. The second factor equals $u^5A(u)+1$ with $A=u^5+t^2u+1$, so its roots are not roots of A; A is separable ($A'=(u+\sqrt{t})^4$, $A(\sqrt{t})=1$) with $e_1=0$; $\ker L_\lambda$ is a 4-dimensional $\mathbb{F}_2$-space containing the u_i, and a relation $u_i+u_{i'}+u_{i''}=0$ would force two of the remaining roots to coincide, while $u_i+u_{i'}+u_{i''}+u_{i'''}=0$ would force the fifth root to be 0; hence the pairwise sums are 10 distinct nonzero kernel elements outside $\{u_i\}$, which

exhausts the kernel. Frobenius commutes with the identification. (c) Rationality over $\mathbb{F}_{q^{12}}$ means $g^{12}=1$. On the sum-zero module $V\cong\mathbb{F}_2[\mathbb{Z}/4]$ of a 4-cycle, $\sum_{i<12}g^i=3(1+g+g^2+g^3)=(1+g)^3$ has rank 1 and $\sum_{i<6}g^{2i}=1+g^2=(1+g)^2$ has kernel of dimension 2; for g of order $1,2,3,6$ both operators vanish on V (the sums have an even number of equal terms), giving trace image 0 and $\ker\cap\operatorname{im}\theta=\ker$. The last sentence follows from the radical dimensions $2k+\dim(\ker\cap\operatorname{im}\theta)$ for W and $\dim(\ker L_\lambda)=4$ for the $S(\lambda_1,\cdot)$, together with the criterion $S\neq 0$ iff $\operatorname{Tr}(\lambda_1\,\cdot)=Q_x$ on $\ker L_\lambda$: for a 4-cycle the functionals $\operatorname{Tr}(\lambda_1\cdot)|_{\ker}$ form a one-dimensional space and $Q_x|_{\ker}$ lies in it (both are determined by whether $\operatorname{Tr}_{q^{12}/q}(x)$ vanishes, as Q_x is additive on the radical and takes the same value on the four roots of degree 4), so exactly half of the λ_1 qualify. □

Proposition 6.8 (the quintic v^5+v+c, k odd)**.** *Let k be odd and $c\in\mathbb{F}_q^\times$. The factorisation type of v^5+v+c over $\mathbb{F}_q$ is one of $(1,4)$, $(2,3)$, $(1,1,1,2)$, and these occur for exactly*

$$N_{(1,4)}=\frac{q-2}{2},\qquad N_{(2,3)}=\frac{q+1}{3},\qquad N_{(1,1,1,2)}=\frac{q-2}{6}$$

values of c. In particular the number of $\mathbb{F}_q$-roots is 1, 0 or 3 with these multiplicities, in agreement with the theorem of Helleseth–Kholosha [12] on $x^{2^l+1}+x+a$, $\gcd(l,k)=1$ (here $l=2$), which rests on Bluher's classification [3] of the roots of $x^{q+1}+ax+b$ (the q of Bluher's title is a power of the characteristic, here 4); and the quintic is never irreducible.

Proof. Put $t=c^{-2/5}$ (a bijection of $\mathbb{F}_q^\times$) and $A(u)=u^5+t^2u+1$, so that $u=t^{1/2}v$ identifies the roots of v^5+v+c with those of A (Proposition 6.7(b)) and $K=\ker L_\lambda\subset\bar{\mathbb{F}}_q$, $\lambda=(\cdot,t,1)$, is the sum-zero $\mathbb{F}_2$-span of the five roots $u_0,\dots,u_4$ of A, on which the q-Frobenius σ acts through its permutation $g\in S_5$ of the roots.

Only three types occur. The function $x\mapsto\operatorname{Tr}_{q/2}(x^5+tx^3)$ on $\mathbb{F}_q$ is an $\mathbb{F}_2$-quadratic form with polar form $\operatorname{Tr}_{q/2}(y\,T_\lambda(x))$, whose radical is $\{x\in\mathbb{F}_q:T_\lambda(x)=0\}=K^\sigma$, the σ-fixed subspace of K. Since the polar form is alternating and nondegenerate on $\mathbb{F}_q/K^\sigma$, $\dim K^\sigma\equiv k\equiv 1\pmod 2$. On the sum-zero module of S_5 the fixed space of g has dimension $1,1,3$ for the cycle types $(1,4),(2,3),(1,1,1,2)$ and $0,2,2,4$ for $(5),(1,1,3),(1,2,2),(1,1,1,1,1)$ (count the g-stable subsets of $\{0,\dots,4\}$ modulo complementation). Hence only the first three types occur.

Counting. (i) The map $v\mapsto v^5+v$ sends $\mathbb{F}_q$ onto the values c, with $v^5+v=0$ iff $v\in\{0,1\}$ (as $v^4=1$ forces $v=1$); so $N_{(1,4)}+3N_{(1,1,1,2)}=q-2$. (ii) Let $v\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$, $\bar v=v^q$, $s=v+\bar v\neq 0$, $n=v\bar v$. Then $\bar v^5+\bar v+v^5+v=(v+\bar v)(s^4+ns^2+n^2+1)$, so $v^5+v\in\mathbb{F}_q$ iff $s^4+ns^2+n^2=1$. Writing $n=s^2m$, irreducibility of v^2+sv+n means $\operatorname{Tr}_{q/2}(m)=1$, and the condition becomes $m^2+m=1+s^{-4}$, which is solvable iff $\operatorname{Tr}_{q/2}(1+s^{-4})=0$ iff $\operatorname{Tr}_{q/2}(s^{-1})=1$ (k odd), in which case exactly one of the two solutions $m,m+1$ has $\operatorname{Tr}m=1$. Hence exactly $2\cdot\#\{s\in\mathbb{F}_q^\times:\operatorname{Tr}(1/s)=1\}=q$ elements $v\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$ satisfy $v^5+v\in\mathbb{F}_q$, necessarily in $\mathbb{F}_q^\times$. A quintic of type $(1,4)$, $(2,3)$, $(1,1,1,2)$ has $0,2,2$ roots in $\mathbb{F}_{q^2}\setminus\mathbb{F}_q$, so $2N_{(2,3)}+2N_{(1,1,1,2)}=q$. (iii) $N_{(1,4)}+N_{(2,3)}+N_{(1,1,1,2)}=q-1$. Solving (i)–(iii) gives the three counts. □

6.6. **The mixed part for odd k: the theorem.** Throughout this subsection k is odd, $(\lambda_3,\lambda_5)=(t,1)$ with $t\in\mathbb{F}_q^\times$ (every class $j=t^5$ has such a representative, Proposition 6.7(a)), $A(u)=u^5+t^2u+1$ with roots $u_0,\dots,u_4$, $K=\ker L_\lambda=\ker T_\lambda\subset\mathbb{F}_{q^{12}}$ (Proposition 6.7(b)), $T=T_\lambda$, σ is the q-Frobenius and $\tau=\operatorname{Tr}_{q^{12}/q}$. For $b\in\mathbb{F}_q$ put

$$S(b)=S(b,t,1)=\sum_{x\in\mathbb{F}_{q^{12}}}\psi_0(x^5+tx^3+bx),\qquad Q_b(x)=\operatorname{Tr}_{q^{12}/2}(x^5+tx^3+bx),$$

so that $W(t,1)=\sum_{b\in\mathbb{F}_q}S(b)$ by (7), and $S(b)\in\{0,\pm 4q^6\}$ since $\dim K=4$.

Lemma 6.9 (shifts)**.** *Let $V=T^{-1}(\mathbb{F}_q)=\{c\in\mathbb{F}_{q^{12}}:T(c)\in\mathbb{F}_q\}$.*

(a) *For $c\in V$ and $b\in\mathbb{F}_q$: $S(b)=(-1)^{Q_b(c)}\,S(b+T(c))$.*
(b) *$c\in V$ iff $(\sigma-1)c\in K$, and then $\tau(c)=(1+\sigma^2+\sigma^4+\cdots+\sigma^{10})\,(\sigma-1)c$.*
(c) *If $c\in V$ and $\tau(c)=0$, then $W(t,1)=(-1)^{Q_0(c)}\,W(t,1)$; in particular $W(t,1)=0$ if $Q_0(c)=1$.*

Proof. (a) $Q_b(x+c) = Q_b(x) + Q_b(c) + \mathrm{Tr}(x\,T(c))$ and $\mathrm{Tr}_{q^{12}/2}(xT(c)) = \mathrm{Tr}_{q^{12}/2}(T(c)x)$ with $T(c) \in \mathbb{F}_q$; substitute $x \mapsto x+c$ in $S(b)$. (b) L_λ has coefficients in $\mathbb{F}_q$ and commutes with σ, so $L_\lambda(c) \in \mathbb{F}_q$ iff $L_\lambda((\sigma-1)c) = 0$; and $T(c) \in \mathbb{F}_q$ iff $T(c)^4 = L_\lambda(c) \in \mathbb{F}_q$. With $c_0 = (\sigma-1)c$ one has $\sigma^i c = c + \sum_{m<i} \sigma^m c_0$, hence $\tau(c) = 12c + \sum_{m\le 10}(11-m)\sigma^m c_0 = \sum_{m \text{ even}} \sigma^m c_0$. (c) Sum (a) over $b \in \mathbb{F}_q$: $Q_b(c) = Q_0(c) + \mathrm{Tr}_{q/2}(b\,\tau(c)) = Q_0(c)$, and $b \mapsto b + T(c)$ permutes $\mathbb{F}_q$. □

Lemma 6.10 (a quadratic factor forces $W = 0$)**.** *Suppose $A(u)$ has an irreducible quadratic factor $u^2 + \kappa u + N$ over $\mathbb{F}_q$ (types $(2,3)$ and $(1,1,1,2)$). Then $N = \kappa^{-3}$, $t = \kappa^2 + \kappa^{-1/2} + \kappa^{-3}$ and $\mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Let u be a root of the quadratic factor and $c \in \mathbb{F}_{q^4}$ with $c^q + c = u$. Then $c \in V$, $\tau(c) = 0$ and $Q_0(c) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Consequently $W(t,1) = 0$. Moreover every $S(b)$, $b \in \mathbb{F}_q$, equals $\pm 4q^6$, and $S(b) = -S(b + T(c))$: the q sums cancel in pairs.*

Proof. Dividing, $A = (u^2 + \kappa u + N)(u^3 + \kappa u^2 + (\kappa^2 + N)u + \kappa^3)$ with $N\kappa^3 = 1$ and $t^2 = \kappa^4 + N\kappa^2 + N^2 = \kappa^4 + \kappa^{-1} + \kappa^{-6}$; the quadratic is irreducible iff $\mathrm{Tr}_{q/2}(N/\kappa^2) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Since $u \in \mathbb{F}_{q^2}$, $\mathrm{Tr}_{q^4/q}(u) = 2(u + u^q) = 0$ and c exists. By Lemma 6.9(b), $(\sigma-1)c = u \in K$ gives $c \in V$, and $\tau(c) = (1 + \sigma^2 + \cdots + \sigma^{10})u = 6u = 0$. The conjugates of c over $\mathbb{F}_q$ are c, $c+u$, $c+\kappa$, $c+u'$ with $u' = u^q = u + \kappa$. As $[\mathbb{F}_{q^{12}} : \mathbb{F}_{q^4}] = 3$ is odd, $Q_0(c) = \mathrm{Tr}_{q^4/2}(c^5 + tc^3) = \mathrm{Tr}_{q/2} \sum_i \big(c_i^5 + tc_i^3\big)$ over the four conjugates $c_i = c + a_i$, $a_i \in \{0, u, \kappa, u'\}$. In characteristic 2, $(c+a)^3 = c^3 + c^2 a + ca^2 + a^3$ and $(c+a)^5 = c^5 + c^4 a + ca^4 + a^5$, and $\sum a_i = \sum a_i^2 = \sum a_i^4 = 0$, so $\sum_i c_i^3 = \sum a_i^3 = u^3 + u'^3 + \kappa^3 = \kappa N = \kappa^{-2}$ and $\sum_i c_i^5 = u^5 + u'^5 + \kappa^5 = \kappa^3 N + \kappa N^2 = 1 + \kappa^{-5}$ (Newton's identities for $u + u' = \kappa$, $uu' = N$). Hence

$$Q_0(c) = \mathrm{Tr}_{q/2}\big(1 + \kappa^{-5} + t\kappa^{-2}\big), \qquad t\kappa^{-2} = 1 + \kappa^{-5/2} + \kappa^{-5},$$

and $\mathrm{Tr}(1) = 1$, $\mathrm{Tr}(\kappa^{-5/2}) = \mathrm{Tr}(\kappa^{-5})$ give $Q_0(c) = \mathrm{Tr}_{q/2}(\kappa^{-5}) = 1$. Lemma 6.9(c) gives $W = 0$. For the last assertion: by Proposition 6.7(c) the trace τ vanishes on K in these types, so $\mathrm{Tr}_{q^{12}/2}(bx) = 0$ for $x \in K$; and Q_0 vanishes on K because it is additive there and each root u_i lies in a subfield $\mathbb{F}_q, \mathbb{F}_{q^2}$ or $\mathbb{F}_{q^3}$ of even index in $\mathbb{F}_{q^{12}}$. By Theorem 6.1, $S(b) \neq 0$ for all b, and Lemma 6.9(a) with $Q_b(c) = 1$ gives the pairing (note $T(c) \notin T(\mathbb{F}_q)$, since otherwise $c \in \mathbb{F}_q + K$ and $Q_0(c)$ would vanish). □

Lemma 6.11 (the 4-cycle: vanishing over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ fixes the sign)**.** *Suppose A has type $(1,4)$: $u_0 \in \mathbb{F}_q$ and σ cycles $u_1 \to u_2 \to u_3 \to u_4 \to u_1$. Put $\kappa = 1 + \mathrm{Tr}_{q/2}(u_0^5 + tu_0^3) \in \mathbb{F}_2$ and, for $b \in \mathbb{F}_q$ and $m \ge 1$, $s_m(b) = \sum_{x \in \mathbb{F}_{q^m}} \psi_0(x^5 + tx^3 + bx)$, so that $S(b) = s_{12}(b)$.*

(a) *$S(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(bu_0) = \kappa$ iff $s_1(b) = 0$; this holds for exactly $q/2$ values of b.*
(b) *If $s_1(b) = 0$ then $s_2(b) = 0$.*
(c) *For these b the L-polynomial of $C_b : y^2 + y = x^5 + tx^3 + bx$ over $\mathbb{F}_q$ is $1 + q^2T^4$, hence $S(b) = +4q^6$. Therefore $W(t,1) = \frac{q}{2} \cdot 4q^6 = 2q^7$.*

Proof. (a) By Theorem 6.1, $S(b) \neq 0$ iff $\mathrm{Tr}_{q^{12}/2}(bx) = Q_0(x)$ for all $x \in K$; both sides are additive on K. For $x = u_0 \in \mathbb{F}_q$ both sides vanish ($\mathrm{Tr}_{q^{12}/q}$ is $12 \cdot \mathrm{id}$ on $\mathbb{F}_q$). For $i \ge 1$, $u_i \in \mathbb{F}_{q^4}$ and $\tau(u_i) = 3\,\mathrm{Tr}_{q^4/q}(u_i) = u_1 + u_2 + u_3 + u_4 = u_0$, so $\mathrm{Tr}_{q^{12}/2}(bu_i) = \mathrm{Tr}_{q/2}(bu_0)$, while $Q_0(u_i) = \mathrm{Tr}_{q/2}\big(\sum_{i' \ge 1}(u_{i'}^5 + tu_{i'}^3)\big) = \mathrm{Tr}_{q/2}\big(p_5 - u_0^5 + t(p_3 - u_0^3)\big)$ with p_m the power sums of all five roots of A; Newton's identities for A ($e_1 = e_2 = e_3 = 0$, $e_4 = t^2$, $e_5 = 1$) give $p_3 = 0$, $p_5 = 5e_5 = 1$, so $Q_0(u_i) = \mathrm{Tr}_{q/2}(1 + u_0^5 + tu_0^3) = \kappa$. Thus $S(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(bu_0) = \kappa$ iff $\mathrm{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 1$. On the other hand $x \mapsto \mathrm{Tr}_{q/2}(x^5 + tx^3 + bx)$ is a quadratic form on $\mathbb{F}_q$ with polar form $\mathrm{Tr}_{q/2}(yT(x))$ and radical $K^\sigma = \{0, u_0\}$, so $s_1(b) \neq 0$ iff $\mathrm{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 0$.

(b) Over $\mathbb{F}_{q^2}$ the radical is $K^{\sigma^2} = \{0, u_0, w, w'\}$ with $w = u_1 + u_3$, $w' = \sigma w = u_2 + u_4 = w + u_0$, and $s_2(b) \neq 0$ iff $\mathrm{Tr}_{q^2/2}(x^5 + tx^3 + bx)$ vanishes at $x = u_0$ and $x = w$. At u_0 it does. Let $N_2 = ww' \in \mathbb{F}_q$ and let $\rho \in \mathbb{F}_q$ be the cube root of u_0 ($3 \nmid q-1$). We claim $N_2 = \rho$. Dividing A by $u + u_0$ and using $u_0^5 + t^2 u_0 + 1 = 0$, the quartic with roots $u_1, \dots, u_4$ is $B(u) = u^4 + u_0 u^3 + u_0^2 u^2 + u_0^3 u + u_0^{-1}$; its resolvent cubic, with roots $u_1u_2 + u_3u_4$, $u_1u_3 + u_2u_4$, $u_1u_4 + u_2u_3$, is $\theta^3 + u_0^2\theta^2 + u_0^4\theta + u_0^6 + u_0 = (\theta + u_0^2)^3 + u_0$, whose unique root in $\mathbb{F}_q$ is $u_0^2 + \rho$; the σ-invariant root is $M = u_1u_3 + u_2u_4$, so $M = u_0^2 + \rho$ and $N_2 = (u_1+u_3)(u_2+u_4) = e_2(B) + M = u_0^2 + M = \rho$.

Now with $w+w'=u_0$, $ww'=\rho$: $w^3+w'^3=u_0^3+u_0\rho$ and $w^5+w'^5=u_0^5+u_0^3\rho+u_0\rho^2$, so

$$\begin{aligned}\mathrm{Tr}_{q^2/2}(w^5+tw^3+bw) &= \mathrm{Tr}_{q/2}(u_0^5+tu_0^3+bu_0)+\mathrm{Tr}_{q/2}\big(u_0^3\rho+u_0\rho^2+tu_0\rho\big)\\ &= 1+\mathrm{Tr}_{q/2}\big(\rho^{10}+\rho^5+t\rho^4\big)=1+\mathrm{Tr}_{q/2}(t\rho^4).\end{aligned}$$

From $u_0^5+t^2u_0+1=0$, $t^2=u_0^4+u_0^{-1}=\rho^{12}+\rho^{-3}$, so $t=\rho^6+\rho^{-3/2}$ and $t\rho^4=\rho^{10}+\rho^{5/2}$, whose trace is $\mathrm{Tr}(\rho^5)+\mathrm{Tr}(\rho^5)=0$. Hence the form takes the value 1 at w and $s_2(b)=0$.

(c) With $\omega_1,\dots,\omega_4$ the Frobenius eigenvalues of C_b (genus 2), $s_m(b)=-\sum_i\omega_i^m$ (Theorem 6.1), so $e_1=\sum\omega_i=-s_1=0$ and $e_2=\frac12\big(e_1^2-\sum\omega_i^2\big)=\frac12(s_1^2+s_2)=0$; the functional equation gives $e_3=qe_1=0$, $e_4=q^2$. So $\prod(1-\omega_iT)=1+q^2T^4$, $\omega_i^4=-q^2$, $\omega_i^{12}=-q^6$ and $S(b)=-\sum\omega_i^{12}=4q^6$. Summing over the $q/2$ values of b in (a) gives $W(t,1)=2q^7$. □

Theorem 6.12 (the mixed part, k odd)**.** *Let k be odd, $\lambda_3\lambda_5\neq0$, $j=\lambda_3^5/\lambda_5^3$ and $c=j^{-1/2}$. Then*

$$W(\lambda_3,\lambda_5)=\begin{cases}2q^7, & v^5+v+c \text{ has type } (1,4),\\ 0, & v^5+v+c \text{ has type } (2,3) \text{ or } (1,1,1,2),\end{cases}$$

and no other type occurs. In the 4-cycle case exactly $q/2$ of the sums $S(\lambda_1,\lambda_3,\lambda_5)$, $\lambda_1\in\mathbb{F}_q$, are nonzero and each equals $+4q^6$; in the other two cases all q sums equal $\pm4q^6$ and cancel in pairs. Equivalently, in the β-form of (7): Q vanishes on the radical $\theta^{-1}(\ker L_\lambda\cap\mathbb{F}_{q^2})$ (of dimension $2k+2$) with Arf sign $+$ in the 4-cycle case, and Q does not vanish on the radical (of dimension $2k+4$) otherwise.

Proof. By Proposition 6.7(a),(b) we may take $(\lambda_3,\lambda_5)=(t,1)$, $t=j^{1/5}$, and the type of v^5+v+c is the type of $A(u)=u^5+t^2u+1$. Proposition 6.8 restricts the types to the three listed (the scaling $(\lambda_1,\lambda_3,\lambda_5)\mapsto(s\lambda_1,s^3\lambda_3,s^5\lambda_5)$ preserves the number and signs of the nonzero S). Lemma 6.10 covers types $(2,3)$ and $(1,1,1,2)$, Lemma 6.11 the type $(1,4)$. For the β-form: by Proposition 6.7(c) the radical has dimension $2k+2$ or $2k+4$, so $|W|\in\{0,2q^7\}$ or $\{0,4q^7\}$ respectively, and $W=2q^7$ (resp. $W=0$) is equivalent to the stated behaviour of Q on the radical. □

Corollary 6.13. *For all odd k: $\sum_{\lambda_3\lambda_5\neq0}W(\lambda_3,\lambda_5)=(q-1)(q-2)q^7$ and*

$$A_6=|U|=q^9+(q-1)q^5.$$

That is, Conjecture 6.5 holds for A_6 whenever k is odd.

Proof. Each class $j\in\mathbb{F}_q^\times$ consists of $q-1$ pairs (λ_3,λ_5) (Proposition 6.7(a)), and the 4-cycle classes number $(q-2)/2$ (Proposition 6.8); Theorem 6.12 gives $\sum_{\text{mixed}}W=(q-1)\frac{q-2}{2}2q^7$, which is (9) for odd k. Combine with Lemmas 6.2, 6.4 and (7). □

Theorem 6.12 was checked in two independent ways: for $k\le7$ by the exact $\mathbb{F}_2$-quadratic-form algorithm (`legendre_d6_qf_mixed.py`: every j, radical test and signs), and for $k\le9$ through the L-polynomials (`kasami_sign.py`): for every $j\in\mathbb{F}_q^\times$ and every $b\in\mathbb{F}_q$ the sums $s_1(b),s_2(b)$ over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ determine $\prod(1-\omega_iT)$ and hence $S(b)=-\sum\omega_i^{12}$; the resulting multisets $\{S(b)\}$ and the values of W agree with the theorem in every case, and in the 4-cycle case the predicate $\mathrm{Tr}_{q/2}(bu_0)=\kappa$ of Lemma 6.11(a) matches $S(b)\neq0$ exactly. The types of Proposition 6.8 were enumerated for $k\le11$ (`quintic_check.py`).

The condition "g is a 4-cycle" is a $\mathbb{Z}/4$-condition (the Frobenius must be a regular unipotent on the kernel), not an $\mathbb{F}_2$-trace condition; this is why no predicate $\mathrm{Tr}(f(j))=0$ describes the set of j with $W\neq0$ (Proposition 6.6). Note also that the sign in the 4-cycle case is *not* obtained from an Arf-invariant computation on $\mathbb{F}_2^{12k}$ but from the two small sums s_1,s_2 over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$, which vanish for exactly the b with $S(b)\neq0$ and force the Frobenius eigenvalues of C_b to be $\sqrt{q}$ times primitive 8th roots of unity. For even k the orbit invariant j is not complete (cube and fifth roots need not exist in $\mathbb{F}_q$) and the analogous analysis has not been carried out; the closed form of X (the sums W') is open for all k.

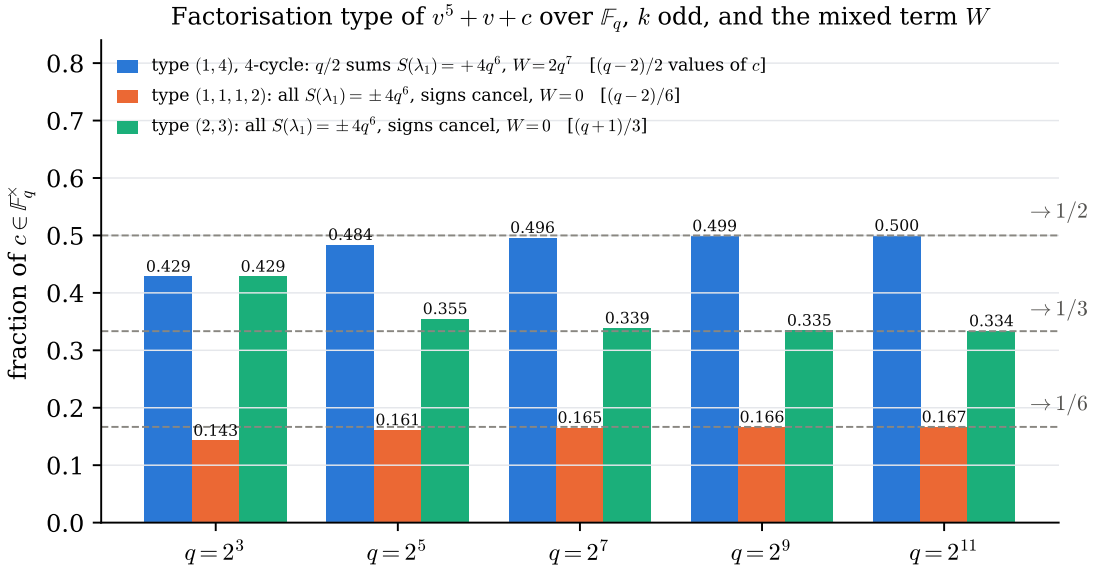


FIGURE 2. The mixed part for odd k (Theorem 6.12). For each $c \in \mathbb{F}_q^\times$ the Gold-type quintic v^5+v+c has one of three factorisation types over $\mathbb{F}_q$, with the exact frequencies of Proposition 6.8 (bars: $q=2^3,\dots,2^{11}$; dashed lines: the limits $\frac{1}{2},\frac{1}{3},\frac{1}{6}$). Only the 4-cycle type contributes to A_6: there $q/2$ of the q curve sums $S(\lambda_1)$ equal $+4q^6$ and the rest vanish; for the other types all q sums are $\pm 4q^6$ and cancel in pairs (Lemmas 6.10, 6.11). The "half-measure" of the 4-cycle set, $(q-2)/2$, is what produces the term $(q-1)(q-2)q^7$ of (9).

6.7. **The sums W' and the constant X.** We now turn to $X = q^{-3}\sum_{\lambda_3,\lambda_5} W'(\lambda_3,\lambda_5)$. Throughout, k is odd unless said otherwise, and we write

$$S'(\lambda_1,\lambda_3,\lambda_5) = \sum_{x\in\mathbb{F}_{q^{12}}} (-1)^{\mathrm{Tr}(\lambda_5 x^5+\lambda_3 x^3+\lambda_1 x)+\mathrm{Tr}_{q/2}\, e_2(x)},$$

$$Q'_\lambda(x) = \mathrm{Tr}(\lambda_5 x^5+\lambda_3 x^3) + \mathrm{Tr}_{q/2}\, e_2(x),$$

where $e_2(x)$ is the second elementary symmetric function of the twelve q-conjugates of x. In Witt-vector terms S' is the sum of a character of order 4 of the Hayes group (the q^3 characters restricting to $(-1)^{\mathrm{Tr}\, c_2}$ on $\mathcal{G}^2$), and $X = q^{-3}\sum_{\lambda\in\mathbb{F}_q^3} S'(\lambda)$.

Proposition 6.14 (structure of W'). *Let $H_0=\{x\in\mathbb{F}_{q^{12}} : \mathrm{Tr}_{q^{12}/q}\, x=0\}$, $L'_\lambda(x)=\lambda_5^4x^{16}+\lambda_3^4x^8+x^4+\lambda_3^2x^2+\lambda_5 x$ (so $L'_\lambda=L_\lambda+x^4$), $K'=\ker L'_\lambda\cap\mathbb{F}_{q^{12}}$, and $P=1+\sigma^2+\sigma^4+\cdots+\sigma^{10}=(\sigma+1)^2(\sigma^2+\sigma+1)^4$. Then for all k:*

(a) *$W'(\lambda_3,\lambda_5)=\sum_{\lambda_1\in\mathbb{F}_q} S'(\lambda) = q\sum_{x\in H_0}(-1)^{Q'_\lambda(x)}$, and Q'_λ is invariant under $x\mapsto x+c$, $c\in\mathbb{F}_q$, on H_0.*

(b) *Q'_λ is an $\mathbb{F}_2$-quadratic form on $\mathbb{F}_{q^{12}}$ with polar form $\mathrm{Tr}\big(y\,(T_\lambda(x)+x+\mathrm{Tr}_{q^{12}/q}\,x)\big)$; on H_0 its radical is $\{x\in H_0 : L'_\lambda(x)\in\mathbb{F}_q\}$, of dimension $k+d$ with $d=\dim_{\mathbb{F}_2}\ker(P|_{K'})$. Hence $|W'(\lambda_3,\lambda_5)|\in\{0,\,2^{d/2}q^7\}$, $d\in\{0,2,4\}$, and the radical of the β-form (7) has dimension $2k+d$.*

(c) *$d=0$ iff $K'=0$, i.e. iff no nonzero root of L'_λ lies in $\mathbb{F}_{q^{12}}$; in that case $W'(\lambda_3,\lambda_5)=\pm q^7$. If $K'\neq 0$ then $d=\dim M_3+\dim\ker(\sigma-1)^2|_{M_1}$, where $K'=M_1\oplus M_3$ is the decomposition into the $(\sigma-1)$- and $(\sigma^2+\sigma+1)$-primary parts.*

Proof. (a) $\sum_{\lambda_1}(-1)^{\mathrm{Tr}(\lambda_1 x)} = q[x\in H_0]$; the β-form follows from $x=\beta+\beta^q$ (q-to-one onto H_0) and $e_2(\beta+\beta^q)=\mathrm{Tr}_{q^{12}/q}(\beta^{q+1})+(\mathrm{Tr}_{q^{12}/q}\,\beta)^2$ (Lemma 5.1). For $c\in\mathbb{F}_q$ and $x\in H_0$: $\mathrm{Tr}(x+c)^5=\mathrm{Tr}\,x^5+c(\mathrm{Tr}\,x)^4+c^4\,\mathrm{Tr}\,x$, $\mathrm{Tr}(x+c)^3=\mathrm{Tr}\,x^3+c(\mathrm{Tr}\,x)^2+c^2\,\mathrm{Tr}\,x$ and $e_2(x+$

$c) = e_2(x) + c\,\mathrm{Tr}\,x + \binom{12}{2}c^2$, all equal to their values at x. (b) The polar form of $\mathrm{Tr}_{q/2}\,e_2$ is $\mathrm{Tr}_{q/2}\big(\mathrm{Tr}\,x\,\mathrm{Tr}\,y + \mathrm{Tr}(xy)\big) = \mathrm{Tr}_{q^{12}/2}\big(y(x+\mathrm{Tr}\,x)\big)$, that of $\mathrm{Tr}(\lambda_5 x^5 + \lambda_3 x^3)$ is $\mathrm{Tr}(yT_\lambda(x))$. On H_0 the term $\mathrm{Tr}\,x$ disappears and $H_0^\perp = \mathbb{F}_q$ for the trace pairing, so the radical is $\{x \in H_0 : T_\lambda(x) + x \in \mathbb{F}_q\} = \{x \in H_0 : L'_\lambda(x) \in \mathbb{F}_q\}$ ($L'_\lambda = (T_\lambda + 1)^4$). As in Lemma 6.9(b), $L'_\lambda(x) \in \mathbb{F}_q$ iff $(\sigma-1)x \in K'$, and $\mathrm{Tr}_{q^{12}/q}\,x = P(\sigma-1)x$; since $(\sigma+1)P = 1 + \sigma + \cdots + \sigma^{11}$, $\ker P \subset H_0$, and the radical is $(\sigma-1)^{-1}(\ker P|_{K'})$, of dimension $k+d$. The exponential sum of a quadratic form on $\mathbb{F}_2^{11k}$ with radical of dimension $k+d$ has modulus 0 or $2^{(12k+d)/2}$, and $W' = q\cdot(\ldots)$. (c) K' is a σ-module with $\sigma^{12} = 1$, so its minimal polynomial divides $x^{12}-1 = (x-1)^4(x^2+x+1)^4$ and has degree ≤ 4; P kills M_3 and acts on M_1 as $(\sigma-1)^2$ times a unit. If $K' \neq 0$ then K' contains a nonzero element killed by $\sigma - 1$ or by $\sigma^2 + \sigma + 1$, so $d > 0$. For $d = 0$ the radical on H_0 is $\mathbb{F}_q$, on which Q'_λ vanishes by (a), so the sum is $\pm 2^{6k}$ and $W' = \pm q^7$. □

Lemma 6.15 (the term $(0,0)$). *For all k, $W'(0,0) = \sum_{\beta\in\mathbb{F}_{q^{12}}}(-1)^{\mathrm{Tr}(\beta^{q+1}+\beta)} = \varepsilon_0\,q^7 = (-1)^{k+1}q^7$.*

Proof. Let $c \in \mathbb{F}_{q^{12}}$ satisfy $c^{q^2} + c = 1$ (solvable, as $\mathrm{Tr}_{q^{12}/q^2}\,1 = 6 = 0$); then $\theta c = c^q + c^{q^{-1}} = \sigma^{-1}(c^{q^2} + c) = 1$. Substituting $\beta \mapsto \beta + c$ and using $\mathrm{Tr}((\beta+c)^{q+1}) = \mathrm{Tr}(\beta^{q+1}) + \mathrm{Tr}(\beta\,\theta c) + \mathrm{Tr}(c^{q+1})$ gives $W'(0,0) = (-1)^{\mathrm{Tr}(c^{q+1}+c)}\sum_\beta(-1)^{\mathrm{Tr}(\beta^{q+1})}$. By Coulter's evaluation [8, Thm. 4.2] of $\sum_{x\in\mathbb{F}_{2^N}}(-1)^{\mathrm{Tr}(x^{2^k+1})}$ with $N = 12k$, $\gcd(k,N) = k$, $N/k = 12$ even and $a = 1$ a (2^k+1)-th power, the last sum equals $-(-1)^{N/2k}2^{N/2+k} = -q^7$. Finally $\mathrm{Tr}_{q^{12}/q^2}\,c = 3c + 3(c+1) = 1$ gives $\mathrm{Tr}_{q^{12}/2}\,c = \mathrm{Tr}_{q^2/2}\,1 = 0$, and $\mathrm{Tr}_{q^{12}/q^2}(c^{q+1}) = 3c\,c^q + 3(c+1)(c^q+1) = c + c^q + 1$, whence $\mathrm{Tr}_{q^{12}/2}(c^{q+1}) = \mathrm{Tr}_{q^2/2}(c + c^q) = \mathrm{Tr}_{q/2}(c + c^{q^2}) = \mathrm{Tr}_{q/2}\,1 = k \bmod 2$. So $W'(0,0) = (-1)^k(-q^7)$. □

Lemma 6.16 (pure cubic part of X, k odd). *Let $\lambda_3 \neq 0$. Then $K' = \{0\}\cup\{z/\lambda_3 : z^3 + z = \lambda_3^2\}$, $d = 2$, and $W'(\lambda_3, 0) \neq 0$ (hence $= \pm 2q^7$) iff the cubic $z^3 + z + \lambda_3^2$ does* not *have exactly one root in $\mathbb{F}_q$, i.e. iff it has type* (3) *or* $(1,1,1)$. *This holds for exactly $q/2$ values of λ_3.*

Proof. $L'_\lambda = \lambda_3^4x^8 + x^4 + \lambda_3^2x^2 = x^2(\lambda_3^2x^3 + x + \lambda_3)^2$, and $x = z/\lambda_3$ turns the cubic into $z^3 + z + \lambda_3^2$. Its roots lie in $\mathbb{F}_q$, $\mathbb{F}_{q^2}$ or $\mathbb{F}_{q^3}$, all of even index in $\mathbb{F}_{q^{12}}$, so $K' = \ker L'_\lambda$ ($\dim 2$), $P = 0$ on K' and $d = 2$. In the β-form the radical is $\theta^{-1}(K')$, $Q' + \mathrm{Tr}$ is additive on it and vanishes on $\theta^{-1}(0) = \mathbb{F}_{q^2}$; for a root $y = z/\lambda_3$ let $\beta_y \in \mathbb{F}_{q^4}$ satisfy $\beta_y^{q^2} + \beta_y = y^q$ (so $\theta\beta_y = y$), and put $v(y) = \mathrm{Tr}_{q^{12}/2}(\lambda_3\beta_y^2 y + \beta_y^{q+1} + \beta_y) = \mathrm{Tr}_{q^4/2}(\lambda_3\beta_y^2 y + \beta_y^{q+1} + \beta_y)$. If $z \in \mathbb{F}_q$, the conjugates of β_y are $\beta_y, \beta_y^q, \beta_y + y, \beta_y^q + y$ and one finds $\mathrm{Tr}_{q^4/q}\,\beta_y = 0$, $\mathrm{Tr}_{q^4/q}(\beta_y^2) = 0$, $\mathrm{Tr}_{q^4/q}(\beta_y^{q+1}) = y^2$, so $v(y) = \mathrm{Tr}_{q/2}(y) = \mathrm{Tr}_{q/2}\big(z^{1/2}/(z+1)\big)$ (from $\lambda_3^2 = z(z+1)^2$) $= \mathrm{Tr}_{q/2}\big(z/(z+1)^2\big) = \mathrm{Tr}_{q/2}\big(\frac{1}{z+1} + \frac{1}{(z+1)^2}\big) = 0$. If $z \in \mathbb{F}_{q^2}\setminus\mathbb{F}_q$ with conjugate $\bar z$, $s = z + \bar z$, $n = z\bar z$, the same computation gives $v(y) = \mathrm{Tr}_{q/2}(s/\lambda_3) + \mathrm{Tr}_{q^2/2}(\lambda_3 y\bar y^2) + \mathrm{Tr}_{q/2}(y\bar y) = \mathrm{Tr}_{q/2}\big(s/\lambda_3 + n(1+s)/\lambda_3^2\big)$; the factorisation $z^3 + z + \lambda_3^2 = (z+s)(z^2 + sz + n)$ gives $n = s^2 + 1$, $\lambda_3^2 = s(s+1)^2$, hence $v(y) = \mathrm{Tr}_{q/2}\big(s^{1/2}/(s+1)\big) + \mathrm{Tr}_{q/2}\big((s+1)/s\big) = 0 + 1 + \mathrm{Tr}_{q/2}(1/s)$, while $z^2 + sz + n$ is irreducible iff $\mathrm{Tr}_{q/2}(n/s^2) = 1 + \mathrm{Tr}_{q/2}(1/s) = 1$. So in type $(1,2)$ the form takes the value 1 on the radical and $W' = 0$; in type $(1,1,1)$ all values vanish; in type (3) the three values are equal and sum to $v(0) = 0$. The count $q/2 = \frac{q+1}{3} + \frac{q-2}{6}$ is the classical distribution of $x^3 + x + c$ for odd k (the same count as in Proposition 6.8, with the roles of the types $(1,2)$ and $(1,4)$). □

Lemma 6.17 (pure quintic part of X: the kernel, k odd). *Let $\lambda_5 \neq 0$, $y_0 = \lambda_5^{-1/5}$ and $c = \lambda_5^2$. Then $K' = \ker L'_\lambda = y_0\{w : w^{15} + y_0^8w^3 + 1 = 0\}\cup\{0\}$, the fifteen elements w are the cube roots $w^3 = \lambda_5^{-2/5}v'$ of the five roots v' of $v'^5 + v' + c$, and the Frobenius orbit structure on $K'\setminus 0$ is $(1,2,4,4,4)$, $(3,6,6)$, $(1,1,1,2,2,2,2,2,2)$ according as $v^5 + v + \lambda_5^2$ has type $(1,4)$, $(2,3)$, $(1,1,1,2)$. Consequently $d = 2$ in the first case and $d = 4$ in the other two, so $|W'(0,\lambda_5)| \in \{0, 2q^7\}$ for $(q-2)/2$ values of λ_5 and $\in \{0, 4q^7\}$ for the remaining $q/2$.*

Proof. With $x = y_0w$ and $\lambda_5y_0^5 = 1$, $y_0^4L'_\lambda(y_0w) = w^{16} + y_0^8w^4 + w$. Put $w^3 = v$: $w^{15} + y_0^8w^3 + 1 = v^5 + y_0^8v + 1$, and $v = y_0^2v'$ gives $v'^5 + v' = y_0^{-10} = \lambda_5^2$. A root $v' \in \mathbb{F}_q$ has one cube root in $\mathbb{F}_q$ and

two in $\mathbb{F}_{q^2}$ ($3 \nmid q-1$); the roots of the irreducible quartic (type $(1,4)$) have their twelve cube roots in $\mathbb{F}_{q^4}$ forming three orbits of length 4; the three roots in $\mathbb{F}_{q^3}$ have one cube root each in $\mathbb{F}_{q^3}$ (one orbit of length 3) and six in $\mathbb{F}_{q^6}$ (one orbit); the two roots in $\mathbb{F}_{q^2}$ have six cube roots forming orbits of length 2 (type $(1,1,1,2)$) or 6 (type $(2,3)$) according to their cubicity in $\mathbb{F}_{q^2}$, which is read off from the orbit count 15. In all cases $K' \subset \mathbb{F}_{q^{12}}$, and d follows from Proposition 6.14(c): the orbit structure $(1,2,4,4,4)$ is a single Jordan block of size 4 (so $\ker(\sigma-1)^2$ has dimension 2), while in the other two cases $(\sigma-1)^2$ or $\sigma^2+\sigma+1$ annihilates everything. □

Proposition 6.18 (structure of the data for W', k odd)**.** *For $k = 3,5,7$ the following hold (`legendre_d6_Wp_pattern.py`; Table 3).*

(a) *Pure terms: $W'(\lambda_3,0) = 2q^7$ whenever it is nonzero (Lemma 6.16), and $W'(0,\lambda_5) = 2q^7$ or $4q^7$ according as $d = 2$ or 4 (Lemma 6.17): $\sum_{\lambda_3\neq 0} W'(\lambda_3,0) = q^8$ and $\sum_{\lambda_5\neq 0} W'(0,\lambda_5) = 3q^8 - 2q^7$.*

(b) *Mixed terms: for every $(\lambda_3,\lambda_5) \in (\mathbb{F}_q^\times)^2$ the Frobenius orbit structure on the 15 nonzero elements of $\ker L'_\lambda$ is one of the eleven patterns obtained from the action of a conjugacy class of $S_6 \cong Sp_4(\mathbb{F}_2)$ on the 15 two-element subsets of $\{1,\dots,6\}$, and the pair (d,W') depends only on the class; the number of pairs in each class is the polynomial in q of Table 3, exactly. The classes of order 7 and 15 of $GL_4(\mathbb{F}_2) \cong A_8$ do not occur.*

(c) *Consequently $\sum_{\lambda_3\lambda_5\neq 0} W' = q^9 - 3q^8 + q^7$ and, with Lemma 6.15 and (a), $\sum_{\lambda_3,\lambda_5} W' = q^9 + q^8$, i.e. $X = q^6 + q^5$.*

TABLE 3. The mixed sums $W'(\lambda_3,\lambda_5)$ for odd k, by the Frobenius class on $\ker L'_\lambda$ (cycle type in S_6; orbit lengths on the 15 nonzero kernel elements), with the radical excess d, the value, and the exact number of pairs $(\lambda_3,\lambda_5) \in (\mathbb{F}_q^\times)^2$ (verified for $k = 3,5,7$). The two classes of order 4 have the same orbit structure and opposite signs; the two classes $2^2 1^2$, 2^3 have the same orbit structure and $W' = 0$, $4q^7$ respectively.

class in S_6	orbits on $\ker L' \setminus 0$	d	W'/q^7	number of pairs
1^6	(1^{15})	4	4	$(q-2)(q-8)/720$
$2\,1^4$	$(1^7,2^4)$	4	0	$(q-2)(q-8)/48$
$2^2\,1^2$	$(1^3,2^6)$	4	0	$q(q-2)/16$
2^3	$(1^3,2^6)$	4	4	$(q-2)(q-8)/48$
$3\,1^3$	$(1^3,3^4)$	4	4	$(q+1)(q-2)/18$
3^2	(3^5)	4	4	$(q+1)(q-2)/18$
$3\,2\,1$	$(1,2,3,3,6)$	4	0	$(q+1)(q-2)/6$
$4\,1^2$	$(1,2,4,4,4)$	2	2	$(q-2)(q-4)/8$
$4\,2$	$(1,2,4,4,4)$	2	-2	$q(q-2)/8$
$5\,1$	$(5,5,5)$	0	-1	$(q^2+1)/5$
6	$(3,6,6)$	4	4	$(q+1)(q-2)/6$

Conjecture 6.19 (X for odd k)**.** *For all odd k: Proposition 6.18(b) holds, in particular $W'(\lambda_3,\lambda_5) = -q^7$ whenever $\ker L'_\lambda \cap \mathbb{F}_{q^{12}} = 0$; the nonzero pure sums are positive; and hence $X = q^6 + q^5$.*

Figure 3 displays the table. Several features of Table 3 are forced by Proposition 6.14: the value of d in every row, the modulus $|W'| \le 2^{d/2}q^7$, and $W' = \pm q^7$ in the class $5\,1$ (where the fifteen roots lie in $\mathbb{F}_{q^5}\setminus\mathbb{F}_{q^{12}}$). The densities of the classes are those of S_6 (1/5 for the 5-cycles, 1/6 for the 6-cycles and for $3\,2\,1$, 1/4 for the two classes of order 4 together, …), so the geometric monodromy group of the family $\{\ker L'_\lambda\}$ over the (λ_3,λ_5)-plane is $Sp_4(\mathbb{F}_2) \cong S_6$ rather than $GL_4(\mathbb{F}_2) \cong A_8$; the exactness of the polynomial counts indicates that, as in Proposition 6.7(b) where $\ker L_\lambda$ is spanned by the five roots of a quintic, $\ker L'_\lambda$ should be the set of pairwise sums of the six roots of a sextic over $\mathbb{F}_q$ with vanishing e_1 (for $\lambda_5 \to 0$ or $\lambda_3 \to 0$ this degenerates to $u\,A_\lambda(u)$ and to the cube-root structure of Lemma 6.17). Exhibiting this sextic, and evaluating

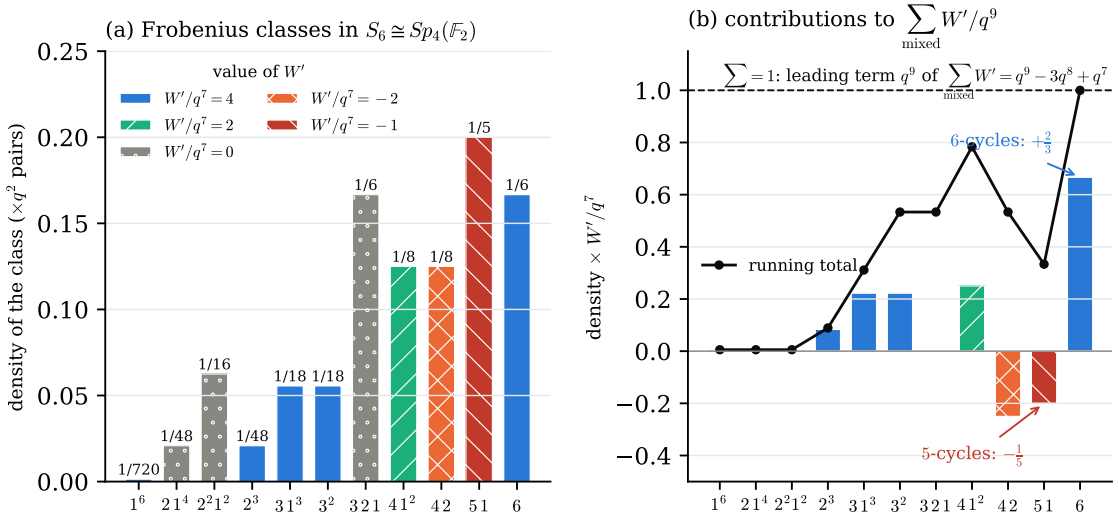


FIGURE 3. The mixed sums W' for odd k by Frobenius class (Table 3), the counterpart of Figure 2. (a) Asymptotic density of each of the eleven classes of $S_6 \cong Sp_4(\mathbb{F}_2)$ among the $(q-1)^2$ pairs (λ_3, λ_5), coloured by the value of W'/q^7: the 5-cycles (1/5) carry $-q^7$; the 6-cycles (1/6), the two classes of order 3 (1/18 each), the class 2^3 and the identity carry $+4q^7$; the two classes of order 4 (1/8 each) carry $\pm 2q^7$ and cancel to leading order; the remaining classes vanish. (b) Density times value: the running total reaches exactly 1, the leading coefficient of $\sum_{\text{mixed}} W' = q^9 - 3q^8 + q^7$ (Proposition 6.18(c)), the negative $-q^9/5$ of the 5-cycles being overcompensated by the 6-cycles ($+2q^9/3$) and the order-3 classes ($+4q^9/9$).

the form on the radical and the Arf invariant class by class as in Lemmas 6.10–6.11, is what a proof of Conjecture 6.19 requires; the sign in the class $5\,1$, which carries the leading term $-q^9/5$, and the signs of the two order-4 classes are the essential points. We have not carried this out.

7. The scaling law at $d = 6$ and the nature of Y

7.1. Variance formula.

Theorem 7.1. *Assume Conjecture 6.5 (true for $q \le 128$; for odd k only its statement for X is needed, by Corollary 6.13, i.e. Conjecture 6.19) and write $Y = q^3 m_k$. Then $B = -2q^3[k \text{ odd}]$,*

$$\varphi_0 = \begin{cases} q^5 + q^4 - 2q^3, & k \text{ odd}, \\ -q^5 + q^4, & k \text{ even}, \end{cases} \qquad \varphi_{A,B} = \begin{cases} -2q^3 \pm q^2 m_k, & k \text{ odd}, \\ \pm q^2 m_k, & k \text{ even}, \end{cases}$$

and

$$\frac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \begin{cases} \dfrac{(q-1)^2(q+2)^2}{q^2} + \dfrac{(q-1)(4q^2+m_k^2)}{q^4}, & k \text{ odd}, \\ (q-1)^2 + \dfrac{(q-1)m_k^2}{q^4}, & k \text{ even}. \end{cases}$$

In particular $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^7 = q^2(1 + O(1/q))$, *i.e.* $C_6 := \lim_{q\to\infty} \mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^9 = 1$, *provided* $m_k = o(q^{5/2})$.

Proof. Substitute into Corollary 4.3. □

Remark 7.2. At $d = 4$ the Legendre anomaly is a single class with relative deficit $1/q$; at $d = 6$ it is the q classes with $c_2 = 0$ (the intervals around f^2 with $f_5 = 0$), each with relative deviation $\pm q^{-2}$ and sign ε_0. Both mechanisms give $\mathrm{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^2$. The full-group variance $\mathrm{Var}_{\mathcal{G}}/q^7$

TABLE 4. $d = 6$: exact variances (Hayes FFT, $q \le 32$; $q = 16$ also by CRT), the three Legendre deviations and $m_k = Y/q^3$. The last two columns show that $\mathrm{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ and not q^4.

q	$\mathrm{Var}_{\mathcal{G}}/q^7$	$\mathrm{Var}_{\mathcal{G}^2}/q^7$	ratio	φ_0	φ_A	φ_B	m_k	$\mathrm{Var}_{\mathcal{G}^2}/q^9$
2	5.7578	15.5625	2.703	32	36	−68	13	3.891
4	3.7004	9.0117	2.435	−768	−16	16	−1	0.563
8	4.7483	79.0935	16.657	35 840	−3 264	1 216	−35	1.236
16	4.3514	225.5056	51.824	−983 040	12 032	−12 032	47	0.881
32	4.2086	1100.8844	261.581	34 537 472	685 056	−816 128	733	1.075

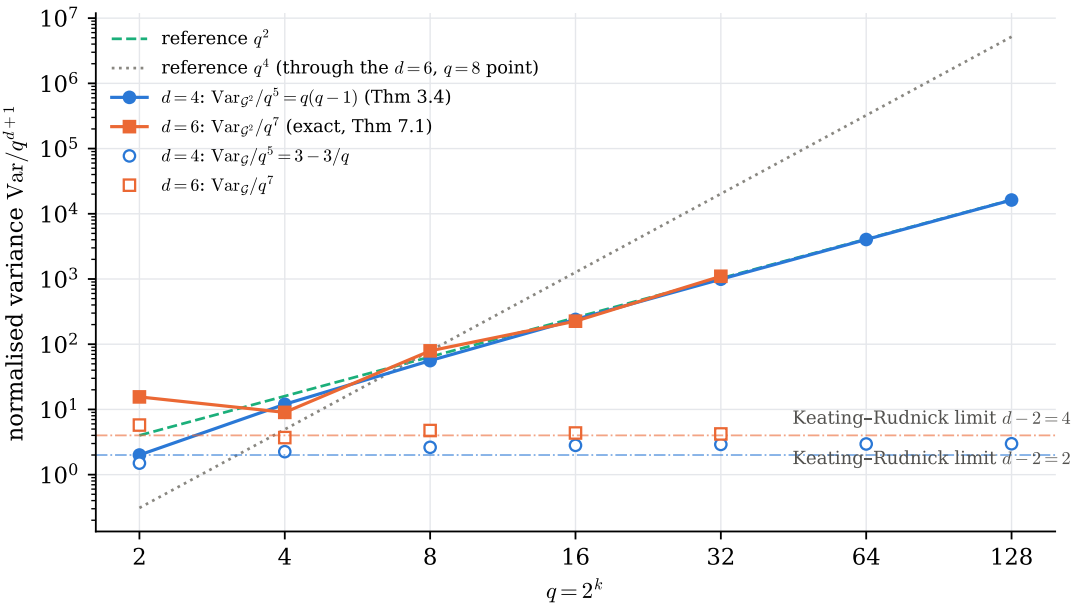


FIGURE 4. Normalised variances against q, both axes logarithmic. Filled markers are $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/q^{d+1}$, exact values ($d = 4$: $q(q-1)$ for $q \le 128$, Theorem 3.4; $d = 6$: Table 4). Open markers: the full-group variances $\mathrm{Var}_{\mathcal{G}}/q^{d+1}$, which approach 3 ($d = 4$, Theorem 3.11) and the Keating–Rudnick value 4 ($d = 6$). The dashed line is q^2; the dotted line is a q^4 law normalised through the $d = 6$, $q = 8$ point, which the data exclude. Whether $\lim \mathrm{Var}_{\mathcal{G}^2}/q^9 = 1$ at $d = 6$ depends on $m_k = o(q^{5/2})$ (Theorem 7.1).

tends to the Keating–Rudnick value 4 (4.35, 4.21 at $q = 16, 32$), unlike $d = 4$ where it tends to $3 \ne 2$. The value $q^{2(d/2-1)} = q^4$ suggested in [5] is excluded.

7.2. **The geometry behind Y: a fixed nine-dimensional variety and a twisted Frobenius.** We now give Y an unambiguous algebro-geometric meaning. The point is that Y_k is a Frobenius trace on a *fixed* variety over $\mathbb{F}_2$, but for a *twisted* Frobenius; in particular $(Y_k)_k$ is not the sequence of $\mathbb{F}_{2^k}$-point counts of any single variety.

Definition 7.3. Let $z = (z_0, \dots, z_{11})$ be coordinates on $\mathbb{A}^{12}_{\mathbb{F}_2}$ and $e_m(z)$ the elementary symmetric polynomials. Put

$$I = \langle e_1, e_3, e_5 \rangle \subset \mathbb{F}_2[z], \qquad X = V(I) \subset \mathbb{A}^{12}_{\mathbb{F}_2}, \qquad f = e_4|_X, \qquad \mathcal{L} = f^*\mathcal{L}_{\psi_0},$$

where $\mathcal{L}_{\psi_0}$ is the Artin–Schreier sheaf on $\mathbb{A}^1_{\mathbb{F}_2}$ attached to $\psi_0(x) = (-1)^x$, so that the trace function of $\mathcal{L} \otimes \mathbb{F}_q$ is $\psi_q \circ e_4$. Let $\sigma \in \mathrm{Aut}(\mathbb{A}^{12})$ be the cyclic shift $\sigma(z)_j = z_{j-1}$ (indices mod 12); it is defined over $\mathbb{F}_2$, preserves I and e_4, and commutes with the absolute Frobenius $F : z \mapsto (z_j^2)$.

The σ-twisted $\mathbb{F}_q$-form $X^{(\sigma)}$ of X is the $\mathbb{F}_q$-structure whose geometric Frobenius is $\sigma \circ F^k$ (the class of a 12-cycle in $H^1(\mathbb{F}_q, S_{12})$; it is the twist defining $\mathrm{Res}_{\mathbb{F}_{q^{12}}/\mathbb{F}_q}\mathbb{A}^1 \cong \mathbb{A}^{12}_{\mathbb{F}_q}$).

Proposition 7.4. (a) *$I = \langle p_1, p_3, p_5\rangle$ with $p_m = \sum_j z_j^m$ (Newton's identities in characteristic 2: $p_3 = e_1^3 + e_1e_2 + e_3$, $p_5 \equiv e_5 + e_2e_3 \pmod{e_1}$).*

(b) *$\alpha \mapsto (\alpha, \alpha^q, \dots, \alpha^{q^{11}})$ is a bijection $U \to X^{(\sigma)}(\mathbb{F}_q) = \{z \in X(\bar{\mathbb{F}}_2) : \sigma F^k(z) = z\}$ carrying $e_4(\alpha)$ to $e_4(z)$; hence $Y = \sum_{z\in X^{(\sigma)}(\mathbb{F}_q)} \psi_q(e_4(z))$.*

(c) *Let $\pi : \mathbb{A}^{12} \to \mathbb{A}^{12}/S_{12} = \operatorname{Spec}\mathbb{F}_2[e_1, \dots, e_{12}]$ and $\Lambda = V(e_1, e_3, e_5) \cong \mathbb{A}^9$ with coordinates $(e_2, e_4, e_6, e_7, \dots, e_{12})$. Then $X = \pi^{-1}(\Lambda)$, $\pi|_X : X \to \Lambda$ is finite flat of degree 12! and generically étale, $\Lambda(\mathbb{F}_q)$ is the set of monic $F \in \mathbb{F}_q[t]$ of degree 12 with $a_{11} = a_9 = a_7 = 0$, and $\#(\pi^{-1}(F) \cap X^{(\sigma)}(\mathbb{F}_q)) = \Lambda(F)$. Thus $Y = \sum_{F\in\Lambda(\mathbb{F}_q)} \Lambda(F)\psi_q(a_8(F))$, which is the Hayes-group form of Section 4; the projection $\Lambda \to \mathbb{A}^2$, (e_2, e_4), has fibres of dimension 7.*

Proof. (a) is immediate from Newton's identities. (b): $\sigma F^k(z)_j = z_{j-1}^q$, so the fixed points are the tuples $z_j = z_0^{q^j}$ with $z_0^{q^{12}} = z_0$, and $e_1 = e_3 = e_5 = 0$ is $p_1 = p_3 = p_5 = 0$ by (a). (c): $\mathbb{F}_2[z]$ is free of rank 12! over $\mathbb{F}_2[e]$; the generic F is separable since $F' = e_7t^4 + e_9t^2 + e_{11} \neq 0$; the σF^k-fixed points of $\pi^{-1}(F)$ are the orderings $z_j = z_0^{q^j}$ of the roots of F, i.e. the $\alpha \in \mathbb{F}_{q^{12}}$ with $\mathrm{charpoly}(\alpha) = F$, of which there are $\Lambda(F)$ (cf. (1)). □

Proposition 7.5 (geometry of X)**.** (a) *X is a complete intersection of codimension 3 in $\mathbb{A}^{12}$ cut out by the regular sequence (e_1, e_3, e_5) of degrees $1, 3, 5$; it is an affine cone of pure Krull dimension 9 and is Cohen–Macaulay. ("Nine-dimensional" refers to $\dim X$; the fibres of $X \to \Lambda$ have dimension 0 and those of $X \to \mathbb{A}^2$ dimension 7.)*

(b) *$\mathrm{Sing}(X) = X \cap \{z : \#\{z_0, \dots, z_{11}\} \le 2\}$ is the union of the 1023 planes $P_A = \{z_j = u\ (j \in A),\ z_j = v\ (j \notin A)\}$, $A \subset \{0, \dots, 11\}$ of even cardinality $0 < |A| < 12$ (A and its complement give the same plane), all containing the diagonal line; $\dim \mathrm{Sing}(X) = 2$, of codimension 7.*

(c) *X is geometrically integral and normal. Consequently the geometric monodromy group of the family $\{F : e_1 = e_3 = e_5 = 0\}$, i.e. the Galois group of $t^{12} + e_2t^{10} + \cdots + e_{11}t + e_{12}$ over $\bar{\mathbb{F}}_2(e_2, e_4, e_6, \dots, e_{12})$, is the full S_{12}.*

Proof. (a) By Proposition 7.4(c), $\mathbb{F}_2[z]/I$ is a free module over the ring $\mathbb{F}_2[e]/(e_1, e_3, e_5)$, which is the polynomial ring $\mathbb{F}_2[e_2, e_4, e_6, \dots, e_{12}]$; so (e_1, e_3, e_5) is a regular sequence and X is finite over $\mathbb{A}^9$. (b) By the Jacobian criterion for complete intersections, X is smooth at z iff de_1, de_3, de_5 are independent there. On X, $\partial_j e_1 = 1$, $\partial_j e_3 = e_2(z \setminus z_j) = e_2 + z_j^2$ and $\partial_j e_5 = e_4(z \setminus z_j) = e_4 + e_2z_j^2 + z_j^4$, so the row space is that of the Vandermonde rows $(1)_j, (z_j^2)_j, (z_j^4)_j$, of rank 3 iff at least three of the z_j are distinct. Conversely, if z takes only the values u, v with multiplicities s, r, $s + r = 12$, then $e_m(z) = \sum_i \binom{s}{m-i}\binom{r}{i}u^{m-i}v^i$; for r even every term with m odd contains a factor $\binom{\text{even}}{\text{odd}}$, which is even by Lucas' theorem, so $e_1 = e_3 = e_5 = 0$ and $P_A \subset X$; for r odd, $e_1 = u + v$ forces $u = v$. (c) X is generically reduced (generic fibre of π étale) and Cohen–Macaulay, hence reduced; Cohen–Macaulay gives Serre's S_2 and (b) gives R_1, so X is normal; a cone is connected, and a connected normal variety is irreducible. The same holds over $\bar{\mathbb{F}}_2$. The irreducible components of $X_{\bar{\mathbb{F}}_2}$ correspond to the cosets S_{12}/G of the geometric monodromy group G (the generic fibre of π is the splitting algebra), so $G = S_{12}$. □

Proposition 7.6 (the Artin–Schreier sheaf)**.** *The cover $Z = \{(z, y) \in X \times \mathbb{A}^1 : y^2 + y = e_4(z)\} \to X$ is étale of degree 2 and Z is smooth over X_{sm}. The function $f = e_4$ has no poles on X; on the projective closure $\overline{X} \subset \mathbb{P}^{12}$ its polar divisor is $4X_\infty$ with $X_\infty \cong \mathbb{P}(X) \subset \mathbb{P}^{11}$ the base of the cone. Since the pole order 4 is divisible by $p = 2$, the Swan conductor of $\mathcal{L}$ along X_∞ is < 4. The sheaf $\mathcal{L}$ is geometrically nontrivial on X (on the diagonal line $e_4 = \binom{12}{4}u^4 = u^4$, and $u^4 \notin \{g^2 + g + c\}$), so Z is geometrically connected and*

$$H_c^{18}(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = H^0(X_{\bar{\mathbb{F}}_2}, \mathcal{L}^\vee)^\vee(-9) = 0.$$

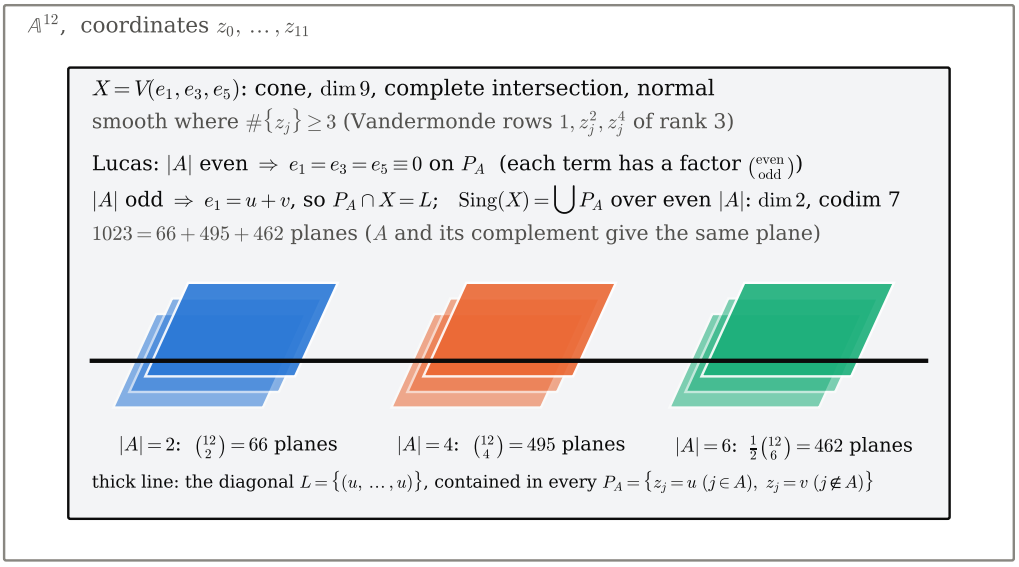


FIGURE 5. The singular locus of $X = V(e_1, e_3, e_5) \subset \mathbb{A}^{12}$ (Proposition 7.5(b)), schematically. A point with only two distinct coordinate values u, v lies on the plane P_A determined by the index set A of the coordinates equal to u; by Lucas' theorem $P_A \subset X$ exactly when $|A|$ is even, and the 1023 such planes, stratified by $|A| \in \{2, 4, 6\}$ (A and its complement give the same plane), all contain the diagonal line and form the two-dimensional singular locus, of codimension 7 in X. Outside these planes at least three coordinates are distinct and the Vandermonde rows $(1), (z_j^2), (z_j^4)$ make X smooth.

Theorem 7.7 (twisted trace formula)**.** *For every $k \geq 1$, with F the geometric Frobenius of $\mathbb{F}_2$ acting on ℓ-adic cohomology ($\ell \neq 2$) and σ acting through its lift to $\mathcal{L}$,*

$$Y_k = \sum_{j=0}^{18} (-1)^j \operatorname{Tr}\big(\sigma \circ F^k \mid H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L})\big).$$

Consequently $\sum_{k \geq 1} Y_k T^k = T \frac{d}{dT} \log L_\sigma(T)$ with $L_\sigma(T) = \prod_j \det(1 - \sigma F T \mid H_c^j)^{(-1)^{j+1}} \in \mathbb{Q}(T)$; $(Y_k)_k$ is a linear recurrence sequence, and $m_k = Y_k / 8^k = \sum_i \pm \zeta_i \eta_i^k$ with $\zeta_i^{12} = 1$ and $\eta_i = \lambda_i / 8$, λ_i the eigenvalues of F. By Deligne [9] the λ_i on H_c^j are algebraic integers of weight $\leq j$; the normalisation $8^k = q^3$ is the Tate twist by 3, so $H_c^j(3)$ has weights $\leq j - 6$. If the cohomology were concentrated in degree $9 = \dim X$ and pure, then $m_k = \pm \sum_{i=1}^{b_9} \zeta_i \eta_i^k$ with $|\eta_i| = 2^{3/2}$ and $|m_k| \leq b_9\, q^{3/2}$.

Proof. By Proposition 7.4(b), Y_k is the sum of the trace function of $\mathcal{L}$ over the $\mathbb{F}_q$-points of the twisted form $X^{(\sigma)}$, whose geometric Frobenius acts on $H_c^\bullet(X_{\bar{\mathbb{F}}_2}, \mathcal{L})$ as σF^k (σ has finite order and commutes with F); apply the Grothendieck–Lefschetz formula (SGA $4\frac{1}{2}$, Rapport; for $F \circ \sigma$ with σ of finite order see also Deligne–Lusztig [10, proof of Thm. 3.2]). Since σ and F commute they are simultaneously triangularisable, which gives the stated shape; L_σ has coefficients in $\mathbb{Q}(\zeta_{12})$ a priori and in $\mathbb{Q}$ because all $Y_k \in \mathbb{Z}$. The weight statements are Weil II. □

By contrast with A_6 and X, the function $(-1)^{\operatorname{Tr} e_4}$ on $\mathcal{G}^2$ is not a character: in Witt coordinates $\mathcal{G}^2 \cong W_2(\mathbb{F}_q)$, $(x_1, x_2) + (x_1', x_2') = (x_1 + x_1', x_2 + x_2' + x_1 x_1')$, and the carry destroys multiplicativity (the characters of order 4 are $i^{\operatorname{Tr} x_1}(-1)^{\operatorname{Tr} x_2 + \cdots}$). Therefore Y does not decompose into Frobenius traces of individual L-functions of curves and no supersingularity argument applies; it is the global twisted trace on the nine-dimensional X.

Proposition 7.8. *The values $m_k = 13, -1, -35, 47, 733$ ($k = 1, \dots, 5$) satisfy $|m_k|/q^{3/2} = 4.60, 0.13, 1.55, 0.73, 4.05$. Hence, in the pure, degree-9 situation of Theorem 7.7, $b_9 \geq 5$. The elementary symmetric functions of the $\zeta_i\eta_i$ computed from $m_1, \dots, m_5$ by Newton's identities are $13, 85, 361, 1069, 2281$; if $b_9 = 5$ then $|e_5| = \prod |\eta_i| = 2^{15/2} \neq 2281$, a contradiction (the argument uses only absolute values, so the twists ζ_i do not affect it). Thus either $b_9 \geq 6$, or $H_c^\bullet$ has contributions outside the pure weight-9 part.*

Proposition 7.4(b) was checked directly: enumerating $X^{(\sigma)}(\mathbb{F}_2)$ inside $\mathbb{F}_{2^{12}}$ gives $|U| = 544$, $Y_1 = 104$, and the pushforward formula of (c) gives $(|U|, Y) = (544, 104)$ and $(259072, -64)$ for $q = 2, 4$ (script `verify_Y_direct.py`; a SageMath version is `verify_Y_sage.sage`).

Proposition 7.9 (Y as a difference of two point counts)**.** *Let $X_c = X \cap \{e_4 = c\}$ ($c \in \mathbb{F}_2$) and $N_c = \#\{\alpha \in U : e_4(\alpha) = c\} = \#X_c^{(\sigma)}(\mathbb{F}_q)$.*

(a) *$N_c = N_1$ for every $c \in \mathbb{F}_q^\times$, $Y = N_0 - N_1$ and $A_6 = N_0 + (q-1)N_1$; hence $N_1 = (A_6 - Y)/q$ and $N_0 = (A_6 + (q-1)Y)/q$. For odd k, by Corollary 6.13,*

$$N_1 = \#X_1^{(\sigma)}(\mathbb{F}_q) = q^8 + q^5 - q^4 - q^2 m_k,$$

so $m_k = O(q^{3/2})$ iff $N_1 = q^8 + q^5 - q^4 + O(q^{7/2})$, and $m_k = o(q^{5/2})$ (the hypothesis of Theorem 7.1 for $C_6 = 1$) iff $N_1 = q^8 + q^5 - q^4 + o(q^{9/2})$.

(b) *X_0 and X_1 are complete intersections of dimension 8 in $\mathbb{A}^{12}_{\mathbb{F}_2}$, stable under σ. The morphism $\varphi : \mathbb{G}_m \times X_1 \to X \setminus X_0$, $(s, z') \mapsto sz'$, satisfies $e_4 \circ \varphi = s^4$ and is finite, surjective and radicial ($s^4 = e_4(z)$ is integral and $s \mapsto s^4$ is injective in characteristic 2), hence a universal homeomorphism; since $\mathcal{L}_{\psi_0}(s^4) \cong \mathcal{L}_{\psi_0}(s)$ and $H_c^\bullet(\mathbb{G}_m, \mathcal{L}_{\psi_0}) = H_c^1$ is one-dimensional of weight 0, there is a long exact sequence of σ- and F-modules*

$$\cdots \to H_c^{j-1}(X_{1,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) \to H_c^j(X_{0,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to H_c^j(X_{1,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) \to \cdots,$$

whose trace formula is (a). In particular the weights of $H_c^j(X, \mathcal{L})$ are bounded by those of $H_c^{j-1}(X_1)$ and $H_c^j(X_0)$.

(c) *(lower vanishing) $H_c^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = 0$ for $j < 9$ and $H_c^j(X_{c,\bar{\mathbb{F}}_2}, \mathbb{Q}_\ell) = 0$ for $j < 8$; dually $H^j(X_{\bar{\mathbb{F}}_2}, \mathcal{L}) = 0$ for $j > 9$. Hence in Theorem 7.7 only $9 \leq j \leq 17$ contribute.*

Proof. (a) U is stable under $\alpha \mapsto s\alpha$ and $e_4(s\alpha) = s^4 e_4(\alpha)$, and $s \mapsto s^4$ is a bijection of $\mathbb{F}_q^\times$; so $Y = \sum_c \chi(c) N_c = N_0 - N_1$ and $A_6 = \sum_c N_c$. (b) $e_4 \notin I$ (on the diagonal line $e_4 = u^4$), and X is integral and Cohen–Macaulay, so $e_1, e_3, e_5, e_4 - c$ is a regular sequence and X_c is a complete intersection of dimension 8. The coordinate ring of $\mathbb{G}_m \times X_1$ is generated over that of $X \setminus X_0$ by s, with $s^4 = e_4$ and $z_i' = z_i s^3/e_4$; φ is bijective on geometric points because $s^4 = e_4(z)$ determines s. A universal homeomorphism induces an equivalence of étale sites, and $s^4 - s = (s^2+s)^2 + (s^2+s)$ gives $\mathcal{L}_{\psi_0}(s^4) \cong \mathcal{L}_{\psi_0}(s)$; Künneth and excision for $X_0 \subset X$ give the sequence. (c) X and X_c are local complete intersections of pure dimensions 9 and 8, so the shifted constant sheaves $\mathbb{Q}_\ell[9]$, $\mathbb{Q}_\ell[8]$ and the lisse rank-one twist $\mathcal{L}[9]$ are perverse; Artin's vanishing theorem for affine varieties [1, 4.1.1] gives $H_c^i(X, \mathcal{L}[9]) = 0$ for $i < 0$ and $H^i(X, \mathcal{L}[9]) = 0$ for $i > 0$. □

Proposition 7.10 (what five values do and do not say)**.** (a) *The minimal linear recurrence over $\mathbb{Q}$ satisfied by $m_1, \dots, m_5$ (Berlekamp–Massey) has order 3, the generic value for five numbers, and is not unique; the recurrences of order 1 and 2 fitted to the first two, resp. four, values predict $m_3 = 1/13$, resp. $m_5 = 727/19$; the characteristic roots of the order-2 fit have modulus $2^{7.48}$, which is not of the form $2^{w/2}$. Five values therefore carry no information about $L_\sigma(T)$ beyond Theorem 7.7.*

(b) *If every eigenvalue of F on $H_c^\bullet(X_{\bar{\mathbb{F}}_2}, \mathcal{L})$ had the form $2^{w/2}\zeta$ with ζ a root of unity (the shape of the eigenvalues behind A_6 and X), then $Y_k = \sum_w 2^{wk/2} A_w^{(k)}$ with algebraic integers $A_w^{(k)}$, and $Y_2 = -2^6$ forces: no single weight $w \geq 7$ can carry Y (so the pure weight-9 case of Proposition 7.8 is impossible for every rank), and a nonzero part of weight ≤ 6 must occur; indeed $v_2(Y_k) = 3k$ exactly for $k \leq 5$ (m_k is odd).*

(c) *For odd k, Corollary 6.13 reads* $\mathrm{Tr}\big(\sigma F^k \mid H_c^\bullet(X_{\bar{\mathbb{F}}_2},\mathbb{Q}_\ell)\big) = 2^{9k}+2^{6k}-2^{5k}$, *and Conjecture 6.5 for A_6 (all k) is equivalent to* $\mathrm{Tr}(\sigma F^k \mid H_c^\bullet(X,\mathbb{Q}_\ell)) = 2^{9k}-(-64)^k+(-32)^k$ *for all k: the untwisted cohomology of X is Tate-like, with σ acting by -1 on the classes of weights* 12 *and* 10.

Proof. (a) is a computation (`L_function_spectral.py`; see Figure 6). (b) $A_w^{(k)}$ is a sum of roots of unity with signs; if $Y_2 = 2^w A$ with A an algebraic integer then $2^{w-6} \mid 1$ in $\bar{\mathbb{Z}}$, so $w \le 6$; and if all weights were ≥ 7 then $v_2(Y_2) \ge 7 > 6$. (c) is Theorem 7.7 with $\mathcal{L}$ replaced by $\mathbb{Q}_\ell$ and $A_6 = \#X^{(\sigma)}(\mathbb{F}_q)$. □

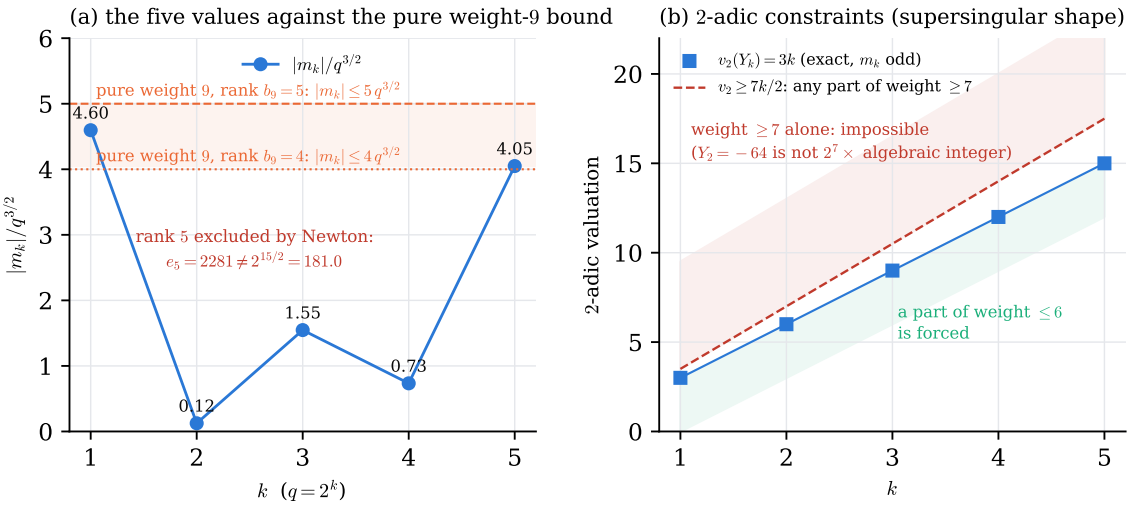


FIGURE 6. What the five values m_k say (Propositions 7.8 and 7.10). (a) $|m_k|/q^{3/2}$ for $k \le 5$ against the bound b_9 of a pure weight-9 object of rank b_9: the value 4.60 at $k=1$ forces $b_9 \ge 5$, and $b_9 = 5$ is excluded because the fifth elementary symmetric function of the normalised eigenvalues computed from $m_1,\dots,m_5$ is $2281 \neq 2^{15/2}$. (b) Under the supersingular shape of the eigenvalues, a part of weight ≥ 7 has 2-adic valuation $\ge 7k/2$, while $v_2(Y_k) = 3k$ exactly (m_k odd); hence no single weight ≥ 7 can carry Y ($Y_2 = -64$) and a part of weight ≤ 6 is forced.

Remark 7.11 (what is and is not established, and a roadmap)**.** Theorem 7.7 and Propositions 7.4–7.6, 7.9–7.10 are unconditional. By Proposition 7.9, $m_k = O(q^{3/2})$ (hence $C_6 = 1$) would follow from: *the eight-dimensional affine complete intersections $X_0 = \{e_1 = e_3 = e_4 = e_5 = 0\}$ and $X_1 = \{e_1 = e_3 = e_5 = 0,\ e_4 = 1\}$ have $H_c^j(X_{c,\bar{\mathbb{F}}_2},\mathbb{Q}_\ell)$ of weight ≤ 9 for $9 \le j \le 15$, and their top classes H_c^{16} cancel in the sequence of* (b). For a smooth affine complete intersection in general position at infinity one expects even more (the cohomology of a bouquet of 8-spheres plus the top class), which would give $Y = O(q^4)$ and $m_k = O(q)$; the data $|m_k|/q \le 23$ do not exclude this for rank ≥ 23. Two obstacles stand in the way: the singular loci (of dimension ≤ 4, where fewer than five coordinates are distinct), and the behaviour at infinity, where the projective closures of X_0 and X_1 meet the hyperplane at infinity in the same variety, the base $\mathbb{P}(X_0)$ of the cone X_0. We have *not* shown that $H_c^j(X_{\bar{\mathbb{F}}_2},\mathcal{L})$ vanishes for $9 < j < 18$ (Proposition 7.6 gives only $j = 18$ and Proposition 7.9(c) only $j < 9$), nor computed b_9: Katz's theorem on the concentration of $\sum \psi(f)$ in middle degree requires $p \nmid \deg f$ and a smooth ambient variety, whereas here $\deg e_4 = 4$ and X is singular along the 1023 planes of Proposition 7.5(b); a proof would need the Swan conductor of $\mathcal{L}$ along X_∞ and local computations at $\mathrm{Sing}(X)$. Hence $m_k = O(q^{3/2})$, and with it $C_6 = 1$, is supported by the data but not proved. Two further cautions. First, the variety $V = \{p_1 = p_3 = p_5 = 0\} \subset \mathbb{A}^{12}_{\mathbb{F}_q}$ obtained by choosing an $\mathbb{F}_q$-basis of $\mathbb{F}_{q^{12}}$ is exactly

$X^{(\sigma)}$, a nontrivial twist of $X \otimes \mathbb{F}_q$; so $(Y_k)_k$ is not a point-count sequence of a fixed variety, and the genuine "closed form" of Y is the twisted L-function $L_\sigma(T)$, which Theorem 7.7 shows is a rational function determined by finitely many Y_k (Berlekamp–Massey on $Y_1, Y_2, \dots$ would identify it; $k \le 5$ is not enough and $k = 6$ means $\mathbb{F}_{2^{72}}$). Second, the untwisted counts $\#X(\mathbb{F}_q)$ for $q = 2, 4, 8$ (2048, 2 099 200, 1 081 081 856) far exceed q^9; this is not a sign of several components but of $q < 12$, where no polynomial over $\mathbb{F}_q$ has twelve distinct roots. The tower reduction of Theorem 5.3 exhibits $Y/G - 1$ as a sum over $a \in \mathbb{F}_{q^6}$ of cubic Artin–Schreier sums over $\mathbb{F}_q$, i.e. as the trace of a relative H^1 of an elliptic Artin–Schreier family over a five-dimensional base; this is the natural starting point for computing b_9.

8. Spectral folding: why the restriction to squares amplifies the variance

At $d = 4$ and $d = 6$ the restriction of Ψ to the subgroup of squares multiplies the normalised variance by a factor $\asymp q^2$ ($q^2/3$ exactly at $d = 4$, $\approx q^2/4$ at $d = 6$), while the full-group variances stay bounded. This section isolates the mechanism in the harmonic analysis of the pair $\mathcal{G}^2 \subset \mathcal{G}$: the variance over a subgroup is the energy of the *folded* spectrum, and folding can only increase the energy through phase coherence inside the fibres of the restriction map $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$.

8.1. **Folding.** Let G be a finite abelian group, $H \le G$, $H^\perp = \{\varepsilon \in \widehat{G} : \varepsilon|_H = 1\} \cong \widehat{G/H}$, and $\varphi : G \to \mathbb{C}$ with Fourier expansion $\varphi = \sum_{\chi \in \widehat{G}} c_\chi \chi$, $c_\chi = |G|^{-1} \sum_g \varphi(g)\bar{\chi}(g)$. Write $\mathrm{Var}_G(\varphi) = |G|^{-1} \sum_{g \in G} |\varphi(g)|^2$ and $\mathrm{Var}_H(\varphi) = |H|^{-1} \sum_{h \in H} |\varphi(h)|^2$ (second moments, not centred on H).

Proposition 8.1 (folding). (a) *The restriction* $\mathrm{Res} : \widehat{G} \to \widehat{H}$ *is surjective and its fibres are the cosets* $\chi H^\perp$, *each of cardinality* $|H^\perp| = [G : H]$.

(b) *(Parseval on G and on H) With* $f_\psi = \sum_{\chi \in \mathrm{Res}^{-1}(\psi)} c_\chi$ *and* $E_\psi = \sum_{\chi \in \mathrm{Res}^{-1}(\psi)} |c_\chi|^2$,

$$\mathrm{Var}_G(\varphi) = \sum_{\psi \in \widehat{H}} E_\psi, \qquad \mathrm{Var}_H(\varphi) = \sum_{\psi \in \widehat{H}} |f_\psi|^2.$$

Moreover, with $F_{gH}(\psi) = |H|^{-1} \sum_{h \in H} \varphi(gh)\bar{\psi}(h)$ *the ψ-coefficient of φ on the coset gH,*

$$f_\psi = F_H(\psi) = |H|^{-1} \sum_{h \in H} \varphi(h)\bar{\psi}(h), \qquad E_\psi = [G : H]^{-1} \sum_{gH \in G/H} |F_{gH}(\psi)|^2.$$

(c) *(Gram matrix) The matrix* $M_{\chi_1,\chi_2} = |H|^{-1} \sum_{h \in H} \chi_1(h)\overline{\chi_2(h)} = [\chi_1\bar{\chi}_2 \in H^\perp]$ *is block diagonal with all-ones blocks indexed by the fibres; its eigenvalues are* $[G : H]$ *(on vectors constant on fibres) and* 0, *and* $\mathrm{Var}_H(\varphi) = c^* M c$, $\mathrm{Var}_G(\varphi) = c^* c$. *Writing* $M = \sum_{\varepsilon \in H^\perp} P_\varepsilon$ *with* P_ε *the translation* $\chi \mapsto \chi\varepsilon$, *the Rayleigh quotient is*

$$\frac{\mathrm{Var}_H(\varphi)}{\mathrm{Var}_G(\varphi)} = 1 + \sum_{\varepsilon \in H^\perp \setminus 1} \rho(\varepsilon), \qquad \rho(\varepsilon) = \frac{\langle c, P_\varepsilon c\rangle}{\langle c, c\rangle},$$

which is (2).

(d) *(coherence bound)* $|f_\psi|^2 \le [G : H]\, E_\psi$ *for every ψ, hence* $\mathrm{Var}_H(\varphi) \le [G : H] \mathrm{Var}_G(\varphi)$, *with equality iff c_χ is constant on every fibre, i.e. iff φ is supported on H. We call* $\kappa_\psi = |f_\psi|^2/([G : H]E_\psi) \in [0, 1]$ *the coherence of the fibre over ψ.*

For the Hayes group $G = \mathcal{G}_\ell$ and $H = \mathcal{G}^2$ one has $[G : H] = |\mathcal{G}[2]| = q^{\lfloor d/2 \rfloor}$ (Lemma 2.1). In odd characteristic $\mathcal{G}_\ell$ has odd order, $\mathcal{G}[2] = 1$, $\mathcal{G}^2 = \mathcal{G}$, and there is nothing to fold: the Legendre intervals are then all the fibres of $\mathcal{G}$, and their variance is the full-group variance of Keating–Rudnick. The amplification is thus a consequence of the 2-torsion of the Hayes group, which exists only in characteristic 2 and has order $q^{d/2}$; the actual factor, $\asymp q^2$ rather than the maximal $q^{d/2}$, measures how much of the spectral energy is coherent along the fibres.

Proof. (a) Characters of H extend to G and two extensions differ by an element of $H^\perp$; $|H^\perp| = |\widehat{G/H}| = [G : H]$. (b) $\varphi|_H = \sum_\psi f_\psi \psi$ and Parseval on H. The projection of φ onto the fibre

$\chi_0 H^\perp$ is

$$\varphi_\psi(g) = \sum_{\chi\in\chi_0 H^\perp} c_\chi \chi(g) = \chi_0(g)\,|H|^{-1} \sum_{h\in H} \varphi(gh)\bar\chi_0(h),$$

so $|\varphi_\psi(g)| = |F_{gH}(\psi)|$ and $E_\psi = |G|^{-1}\sum_g |\varphi_\psi(g)|^2$. (c) Orthogonality of characters on H; $c^*Mc = \sum_\psi |\sum_{\text{fibre}} c_\chi|^2$. (d) Cauchy–Schwarz on each fibre; equality iff $c_{\chi\varepsilon} = c_\chi$ for all $\varepsilon \in H^\perp$ iff $\varphi(g)\varepsilon(g) = \varphi(g)$ for all g and $\varepsilon \in H^\perp$ iff φ vanishes off H. □

8.2. **The case $d = 4$: a single class folded q^2 times.**

Theorem 8.2. *Let $d = 4$, $\varphi = \Psi - q^5$ on $\mathcal{G}_3$, $H = \mathcal{G}^2 = \{(0,c,0)\} \cong \mathbb{F}_q$ with characters $\psi_\mu(c) = \chi(\mu c)$. Then the fibre sums are*

$$f_{\psi_0} = \big((-1)^k - 1\big)(q^3 - q^2), \qquad f_{\psi_\mu} = -q^3 + q^2 - (-1)^k q^2 \quad (\mu \neq 0),$$

so that $\mathrm{Var}_{\mathcal{G}^2} = \sum_\mu |f_{\psi_\mu}|^2 = q^7 - q^6$, while $\mathrm{Var}_{\mathcal{G}} = \sum_\mu E_{\psi_\mu} = 3q^5 - 3q^4$, of which the identity class alone contributes $(q^4 - q^3)^2/q^3 = q^5 - 2q^4 + q^3$. Hence

$$\frac{\mathrm{Var}_{\mathcal{G}^2}}{\mathrm{Var}_{\mathcal{G}}} = \frac{q^2}{3}, \qquad \frac{\mathrm{Var}_{\mathcal{G}}}{q^5} = \underbrace{1 - \frac{2}{q} + \frac{1}{q^2}}_{\text{identity class}} + \underbrace{2 - \frac{1}{q} - \frac{1}{q^2}}_{\text{all other classes}},$$

the second bracket tending to the Keating–Rudnick value $d - 2 = 2$. The fibre over ψ_μ ($\mu \neq 0$) consists of the q^2 characters χ of $\mathcal{G}_3$ with $\chi^2 = \varepsilon_{(\sqrt{\mu},0)}$ (order 4), the fibre over ψ_0 of the q^2 quadratic characters ε_λ with $c_{\varepsilon_\lambda} = S_\lambda/q^3$, $S_\lambda = \sum_{x\in\mathbb{F}_{q^8}} \psi_0(\lambda_3 x^3 + \lambda_1 x)$.

Proof. By Theorem 3.2, $\varphi(0,0,0) = -q^4 + q^3$ and $\varphi(0,c,0) = (-1)^k q^3$ for $c \neq 0$; so $f_{\psi_\mu} = q^{-1}\sum_c \varphi(0,c,0)\chi(\mu c) = (-1)^k q^3[\mu = 0] + (-q^4 + q^3 - (-1)^k q^3)/q$. The sum of squares is $q^6(q-1)$ in both parities, and Theorem 3.11 gives $\mathrm{Var}_{\mathcal{G}}$. For $g \in \mathcal{G}_3$, $\chi(g^2) = \chi(g)^2$ and $c_2(g^2) = c_1(g)^2$, so $\chi|_{\mathcal{G}^2} = \psi_\mu$ iff $\chi^2 = \varepsilon_{(\sqrt{\mu},0)}$; for $\mu = 0$ these are the characters trivial on $\mathcal{G}^2$, i.e. the ε_λ of Proposition 2.2, whose coefficients are the Artin–Schreier sums of Section 3. □

Thus at $d = 4$ the whole anomaly is the identity class: its deficit $-(q^4 - q^3)$ (the missing $\alpha \in \mathbb{F}_{q^8}$ with $e_1 = e_2 = e_3 = 0$) has a flat Fourier transform on $\widehat{\mathcal{G}^2}$, $f_{\psi_\mu} \approx -q^3$ for every μ, so every fibre is coherent (Table 5: $\kappa \approx 1/3$); on the full group the same class is diluted by the factor $[\mathcal{G} : \mathcal{G}^2] = q^2$ and contributes only the "1" in $\mathrm{Var}_{\mathcal{G}}/q^5 \to 3 = 2 + 1$. Theorem 3.12 is the same statement in the language of ρ: $\rho(\varepsilon) = 1$ exactly for the ε trivial on $\{c_1 = 0\} \supset \operatorname{supp}\varphi$ (translation by such ε fixes the coefficient vector), and the remaining $\rho(\varepsilon) = (q-3)/(3(q-1)) \to 1/3$ is the coherence along the λ_3-direction.

8.3. **The case $d = 6$: q coherent fibres out of q^2.**

Proposition 8.3. *Let $d = 6$, $H = \mathcal{G}^2 = \{(c_2, c_4)\} \cong W_2(\mathbb{F}_q)$ with the Witt law $(c_2,c_4)(c_2',c_4') = (c_2 + c_2', c_4 + c_4' + c_2c_2')$, whose q^2 characters are $\psi_{(a,b)}(x) = i^{\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}((a,b)\cdot x)}$: $\psi_{(0,b)} = \chi(\sqrt{b}\,c_2)$ has order ≤ 2 (q characters) and $\psi_{(a,b)}$, $a \neq 0$, has order 4 ($q^2 - q$ characters). Then, for all k,*

$$f_{\psi_{(0,0)}} = \frac{A_6 - q^9}{q^2}, \qquad f_{\psi_{(0,b)}} = \frac{X}{q^2}\ (b \neq 0), \qquad \sum_{a\neq 0,\, b} |f_{\psi_{(a,b)}}|^2 = q^3(q-1)m_k^2,$$

and consequently

$$\mathrm{Var}_{\mathcal{G}^2}(\Psi) = \frac{(A_6 - q^9)^2 + (q-1)X^2}{q^4} + q^3(q-1)\,m_k^2. \tag{10}$$

The fibre over $\psi_{(0,0)}$ consists of the q^3 quadratic characters ε_λ (coefficients $S(\lambda)/q^5$, Theorem 6.1), the fibre over $\psi_{(0,b)}$ of the q^3 characters of order 4 with $\chi^2 = \varepsilon_{(b^{1/4},0,0)}$ (coefficients $S'(\lambda)/q^5$ up to the scaling b, §6.7), and the fibres over the order-4 characters of $\mathcal{G}^2$ of characters of order 8 of $\mathcal{G}_5$, whose coefficients are Witt-vector sums of length 3 and are not quadratic-form sums.

Proof. $f_\psi = q^{-2}\sum_{g\in\mathcal{G}^2}\varphi(g)\bar\psi(g)$. For $\psi=\psi_{(0,b)}$ this is $q^{-2}\big(\sum_{\mathcal{G}^2}\Psi(g)\chi(\sqrt{b}\,c_2) - q^7\cdot q^2[b=0]\big)$, i.e. $(A_6-q^9)/q^2$ or $X_{\sqrt{b}}/q^2 = X/q^2$ (scaling $\alpha\mapsto s\alpha$, $e_2\mapsto s^2e_2$). By Theorem 4.2, $\varphi|_{\mathcal{G}^2} = \varphi_0[c_2=0] + (B + q^2 m_k\,\chi(c_4/c_2^2))[c_2\neq 0]$; the first two terms are functions of c_2 alone, hence combinations of the $\psi_{(0,b)}$, while $\sum_{c_4}\chi(c_4/c_2^2)\bar\psi_{(0,b)}(c_2,c_4)=0$ for $c_2\neq 0$, so the last term is supported on the order-4 characters, where Parseval gives $q^{-2}\sum_{c_2\neq0,c_4}(q^2m_k)^2 = q^3(q-1)m_k^2$. The description of the fibres follows from $\chi(g^2)=\chi(g)^2$ and $c_2(g^2)=c_1(g)^2$ as at $d=4$ (characters with $\chi^2=\varepsilon$ of order 2 have order 4; those restricting to an order-4 character of $\mathcal{G}^2$ have order 8). □

Formula (10) separates the variance into a *coherent* part carried by the q fibres over the characters of order ≤ 2, which is determined by the two constants A_6 and X, and an *incoherent* part $q^3(q-1)m_k^2$ carried by the q^2-q fibres over the order-4 characters, which is Y. With $A_6 = q^9+(q-1)q^5$ (Corollary 6.13) and $X=q^6+q^5$ (Conjecture 6.19) the coherent part is $(q-1)(q+3)q^7$, and the three-value variance of Theorem 7.1 is recovered. The coherent fibres have sums $\approx q^4$, which is q^3 coefficients of typical size q added with the same sign: the order-4 sums $S'(\lambda)$ of §6.7 take the values $0,\pm q^6,\pm 2q^6,\pm 4q^6$ with the strong positive bias displayed in Table 3, and the quadratic sums $S(\lambda)$ are $+4q^6$ on the 4-cycle classes of Theorem 6.12. This Arf-invariant bias of the supersingular quadratic forms is the arithmetic source of the coherence; the incoherent part, by contrast, involves no quadratic form and stays of the size $q^3(q-1)m_k^2$, negligible against q^9 as long as $m_k = o(q^{5/2})$. Since $\mathrm{Var}_{\mathcal{G}}/q^7\to 4$ (Table 4), the amplification is $\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}\approx q^2/4$: q coherent fibres with coherence $\kappa\approx 1/10$ out of $|\mathcal{G}[2]| = q^3$ possible, carrying about $1/7$ of the total energy at $q=16$ (Table 5 and Figure 7).

TABLE 5. Spectral folding data (`fourier_spectrum_folding.py`): the fibres of $\widehat{\mathcal{G}}\to\widehat{\mathcal{G}^2}$ grouped by the order of the character of $\mathcal{G}^2$; "share" is the fraction of $\mathrm{Var}_{\mathcal{G}^2}$, resp. $\mathrm{Var}_{\mathcal{G}}$, carried by the group of fibres, and κ the coherence of Proposition 8.1(d). At $d=4$ all classes come from Theorem 3.2; at $d=6$ from the Hayes FFT.

d	q	$\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}$	$[\mathcal{G}:\mathcal{G}^2]$	order of ψ	#fibres	share of $\mathrm{Var}_{\mathcal{G}^2}$ / $\mathrm{Var}_{\mathcal{G}}$	κ
4	8	$21.33 = q^2/3$	64	1	1	0.438 / 0.167	0.875
				2	7	0.563 / 0.833	0.225
4	32	$341.33 = q^2/3$	1024	1	1	0.121 / 0.042	0.969
				2	31	0.879 / 0.958	0.306
6	8	$16.66 = 0.260\,q^2$	512	1	1	0.077 / 0.036	0.070
				2	7	0.896 / 0.228	0.128
				4	56	0.027 / 0.736	0.001
6	16	$51.82 = 0.202\,q^2$	4096	1	1	0.062 / 0.008	0.095
				2	15	0.935 / 0.138	0.086
				4	240	0.002 / 0.854	$3\cdot 10^{-5}$

Remark 8.4. The two cases are instances of one pattern. In both, the deviation φ is large on a small subgroup $K\subset\mathcal{G}^2$ ($K=\{1\}$ at $d=4$ with $\varphi\approx -q^4$; $K=\{c_2=0\}$ of order q at $d=6$ with $\varphi\approx\varepsilon_0 q^5$), and the Fourier transform of a function supported on K is constant along the fibres of $\widehat{\mathcal{G}^2}\to\widehat{K}$. Folding to $\mathcal{G}^2$ therefore adds these coefficients in phase, whereas on $\mathcal{G}$ they are spread over $[\mathcal{G}:\mathcal{G}^2]$ times more classes. The size of φ on K is in turn a statement about the number of $\alpha\in\mathbb{F}_{q^n}$ with prescribed vanishing of symmetric functions ($e_1=e_2=e_3=0$ at $d=4$; $e_1=e_2=e_3=e_5=0$ at $d=6$, Proposition 7.9), i.e. about the point counts of the varieties X, X_0, X_1: the "q^2" is $|K|\varphi_K^2/|\mathcal{G}^2|$ divided by q^{d+1}, which equals q^2 in both cases ($1\cdot q^8/q$ and $q\cdot q^{10}/q^2$ against q^5 and q^7). What is specific to characteristic 2 is not the folding identity but the existence of the subgroup $\mathcal{G}^2$ of index $q^{d/2}$ and the Arf-invariant bias that makes φ_K as large as q^d, resp. q^{d-1}.

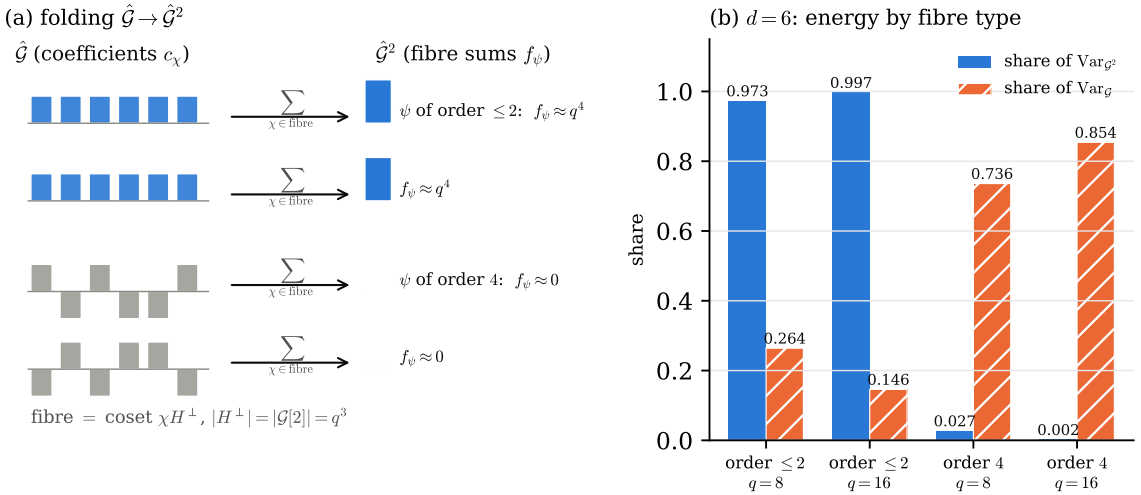


FIGURE 7. Spectral folding at $d = 6$. (a) The restriction map $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$: each fibre is a coset of $H^\perp = \widehat{\mathcal{G}/\mathcal{G}^2}$ of size $|\mathcal{G}[2]| = q^3$, and the fibre sum f_ψ is the ψ-coefficient of φ on $\mathcal{G}^2$. Over the q characters of $\mathcal{G}^2$ of order ≤ 2 the coefficients add in phase (they are the biased supersingular sums S, S' of Sections 6.6–6.7); over the $q^2 - q$ characters of order 4 they cancel. (b) Shares of $\mathrm{Var}_{\mathcal{G}^2}$ and of $\mathrm{Var}_{\mathcal{G}}$ carried by the two kinds of fibres for $q = 8, 16$ (Table 5): the coherent fibres hold 97–100% of the Legendre variance but only 15–26% of the full-group energy.

9. Discussion, open problems, and general even d

9.1. **Translation orbits for general even d.** The symmetry that produced the three-value profile at $d = 6$ exists for every even d, and Lucas' theorem decides how it acts on the identity class.

Proposition 9.1. *Let d be even, $n = 2d$, $\ell = d - 1$, and $U_d = \{\alpha \in \mathbb{F}_{q^n} : \operatorname{Tr} \alpha^j = 0$ for odd $j \le \ell\}$, the set of α whose class lies in $\mathcal{G}^2$.*

(a) *For $c \in \mathbb{F}_q$ the translation $\alpha \mapsto \alpha + c$ preserves U_d and induces a bijection τ_c of $\mathcal{G}^2$ with $\Psi \circ \tau_c = \Psi$; explicitly $e_j(\alpha + c) = \sum_{i \le j} \binom{n-i}{j-i} e_i(\alpha) c^{j-i}$, so τ_c depends only on the binomial coefficients $\binom{2d-i}{j-i} \bmod 2$, $i \le j \le d - 1$.*

(b) *The orbit of the identity class is $K_d = \{(1 + cu)^{2d} \bmod u^d : c \in \mathbb{F}_q\} = \{(\binom{2d}{j} c^j)_{j \le d-1} : c \in \mathbb{F}_q\}$, and Ψ is constant on K_d. By Lucas' theorem $\binom{2d}{j}$ is odd iff the binary digits of j are among those of $2d$; hence $|K_d| = 1$ if d is a power of 2, and $|K_d| = q$ otherwise, the first nonzero coordinate of K_d being $c^{2^{v+1}}$ in position 2^{v+1} where $2^v \parallel d$. In particular $K_4 = \{1\}$, $K_6 = \{c_2 = 0\} = \{1 + c^4 u^4\}$ (Section 4), $K_8 = \{1\}$, $K_{10} = \{1 + c^4 u^4\}$, $K_{12} = \{1 + c^8 u^8\}$, $K_{14} = \{1 + c^4 u^4 + c^8 u^8 + c^{12} u^{12}\}$.*

(c) *(lower bound) $\mathrm{Var}_{\mathcal{G}^2}(\Psi) \ge |K_d|\, \varphi(1)^2 / |\mathcal{G}^2|$ with $|\mathcal{G}^2| = q^{d/2-1}$, where $\varphi(1) = \Psi(1) - q^{d+1} = \#\{\alpha \in \mathbb{F}_{q^{2d}} : e_1(\alpha) = \cdots = e_{d-1}(\alpha) = 0\} - q^{d+1}$ is the twisted point count of the complete intersection $V_d = V(e_1, \ldots, e_{d-1}) \subset \mathbb{A}^{2d}_{\mathbb{F}_2}$ of dimension $d + 1$, minus its main term. Hence $\mathrm{Var}_{\mathcal{G}^2} / q^{d+1} \to \infty$ as soon as $|\varphi(1)| / q^{3d/4} \to \infty$.*

Proof. (a) $\operatorname{Tr}(\alpha + c)^j = \sum_i \binom{j}{i} c^{j-i} \operatorname{Tr} \alpha^i$ and on U_d all $\operatorname{Tr} \alpha^i$ with $i < j$ vanish (odd i by definition, even i because $\operatorname{Tr} \alpha^{2i} = (\operatorname{Tr} \alpha^i)^2$), while $\binom{j}{0} c^j \operatorname{Tr} 1 = n c^j = 0$; the formula for $e_j(\alpha + c)$ is the expansion of $\prod_i (t - c - \alpha_i)$, and it shows that $(e_i(\alpha))_{i \le j}$ determines $(e_i(\alpha + c))_{i \le j}$, so τ_c is well defined on classes; it is bijective with inverse $\tau_{-c} = \tau_c$, and $\Psi \circ \tau_c = \Psi$ because $\alpha \mapsto \alpha + c$ is a bijection of $\mathbb{F}_{q^n}$ compatible with the classes. (b) For α in the identity class only the term $i = 0$ survives, giving $e_j(\alpha + c) = \binom{n}{j} c^j$, i.e. the class of $(1 + cu)^n$. Lucas' theorem gives the parity of

$\binom{2d}{j}$; the least $j \geq 1$ with $\binom{2d}{j}$ odd is the lowest set bit 2^{v+1} of $2d$, which is $\leq d-1$ iff d is not a power of 2; if it is $\leq d-1$, then $c \mapsto c^{2^{v+1}}$ is injective and $|K_d| = q$. (c) is the contribution of K_d to the second moment. □

9.2. **The variance for $d = 8$ and $d = 10$, and what is and is not universal.** Table 6 collects the exact Legendre variances for $d \leq 10$ (Hayes FFT, `general_d_scan.py`; the $d \leq 7$ values agree with [5, Table 4]). Three facts stand out (Figure 8).

TABLE 6. Legendre and full-group variances for even $d \leq 10$ (second moments about q^{d+1}), the identity-class deviation $\varphi(1)$, the orbit K_d, and the share of $\sum_{\mathcal{G}^2} \varphi^2$ carried by K_d.

d	q	$\mathrm{Var}_{\mathcal{G}}/q^{d+1}$	$\mathrm{Var}_{\mathcal{G}^2}/q^{d+1}$	ratio	ratio$/q^2$	$\varphi(1)/q^{d-1}$	$\lvert K_d\rvert$	share of K_d
4	4	2.250	12.00	5.33	0.333	−3.000	1	0.750
4	8	2.625	56.00	21.33	0.333	−7.000	1	0.875
4	32	2.906	992.00	341.33	0.333	−31.000	1	0.969
6	4	3.700	9.01	2.44	0.152	−0.750	4	0.999
6	8	4.748	79.09	16.66	0.260	+1.094	8	0.968
6	32	4.209	1100.88	261.58	0.255	+1.029	32	0.986
8	2	4.881	1.59	0.33	0.082	−0.313	1	0.245
8	4	5.878	13.25	2.25	0.141	−0.387	1	0.181
8	8	5.927	13.49	2.28	0.036	−0.234	1	0.260
10	2	7.192	7.91	1.10	0.275	+0.094	2	0.018
10	4	7.693	10.47	1.36	0.085	−0.076	4	0.140

First, the full-group variance follows Keating–Rudnick at every d ($\mathrm{Var}_{\mathcal{G}}/q^{d+1} \to d-2$, with the exact deviation $1 - 3/q$ at $d = 4$ coming from the identity class, Theorem 8.2). Second, the q^2-amplification of the Legendre variance is a feature of $d = 4$ and $d = 6$: there the identity orbit K_d carries essentially all of $\sum_{\mathcal{G}^2} \varphi^2$, because $\varphi(1)$ is of order q^d (exactly $-q^4 + q^3$) resp. q^{d-1} (exactly $\varepsilon_0 q^5 + q^4 - 2q^3$ for odd k), far above the threshold $q^{3d/4}$ of Proposition 9.1(c). Third, at $d = 8$ the amplification *disappears*: $\mathrm{Var}_{\mathcal{G}^2}/q^9$ is 13.25 and 13.49 at $q = 4, 8$ and the ratio to the full-group variance is 2.25, 2.28, with the identity class ($|K_8| = 1$) carrying only a quarter of the Legendre second moment and $\varphi(1) \approx -1.9\, q^6 = O(q^{3d/4})$ exactly at the threshold; at $d = 10$ the ratio is 1.10, 1.36 for $q = 2, 4$. This refutes the law $\mathrm{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^{2(d/2-1)}$ suggested in [5] (it would give q^6 at $d = 8$), and it equally refutes the natural guess that the q^2 of $d = 4, 6$ is universal. In the language of Section 8: at $d = 8$ the index $[\mathcal{G} : \mathcal{G}^2] = q^4$ is as large as ever, but the deviation is no longer concentrated on a small subgroup, so the fibres of $\widehat{\mathcal{G}} \to \widehat{\mathcal{G}^2}$ are incoherent and the Rayleigh quotient stays bounded.

Conjecture 9.2 (general even d)**.** *Let $d \geq 4$ be even and $q = 2^k \to \infty$. Then $\mathrm{Var}_{\mathcal{G}}(\Psi)/q^{d+1} \to d-2$, and the ratio $\mathrm{Var}_{\mathcal{G}^2}(\Psi)/\mathrm{Var}_{\mathcal{G}}(\Psi)$ is $\asymp q^2$ for $d = 4$ and $d = 6$ and bounded for $d = 8$ and $d = 10$. More precisely, the ratio is unbounded iff the identity-class deviation satisfies $|\varphi(1)|/q^{3d/4} \to \infty$, i.e. iff the twisted point count of $V_d = V(e_1, \ldots, e_{d-1})$ has a subleading term of weight $> 3d/2$ with nonzero σ-twisted trace; this holds for $d = 4, 6$ (weights $8 = 2d$ and $10 = 2(d-1)$) and fails for $d = 8$.*

The first assertion is the Keating–Rudnick asymptotic, which in characteristic 2 rests on Katz's equidistribution theorem [14] and is not at issue here; the equivalence in the last sentence is Proposition 9.1(c) in one direction, and in the other direction it asserts that the classes outside K_d never produce an amplification by themselves, which is what Table 6 shows for $d \leq 10$ and what the folding picture of Section 8 predicts (coherence requires a deviation concentrated on a subgroup, and the only subgroup singled out by the symmetries is K_d). We have no proof of either direction beyond $d = 6$.

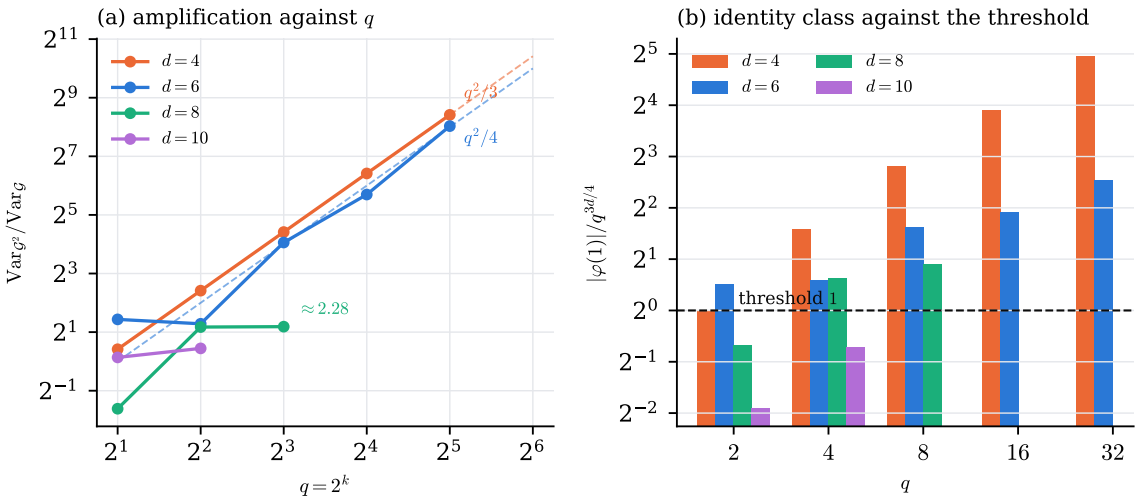


FIGURE 8. The Legendre amplification for even d (data of Table 6). (a) $\mathrm{Var}_{\mathcal{G}^2}/\mathrm{Var}_{\mathcal{G}}$ against q on logarithmic scales: $q^2/3$ exactly at $d=4$, about $q^2/4$ at $d=6$, and no growth at $d=8$ (2.25, 2.28 at $q=4,8$) or $d=10$. (b) The identity-class deviation $|\varphi(1)|$ relative to the threshold $q^{3d/4}$ of Proposition 9.1(c): far above it for $d=4$ (where it equals $q-1$) and $d=6$, at it for $d=8$ (1.5, 1.9 at $q=4,8$), below it for $d=10$.

9.3. **Open problems.**

Open Problem 9.1. (*the constant X*) Prove Conjecture 6.19: exhibit the sextic over $\mathbb{F}_q$ with $e_1=0$ whose pairwise root sums form $\ker L'_\lambda$, so that the Frobenius acts on $\ker L'_\lambda$ through $Sp_4(\mathbb{F}_2)\cong S_6$ with the class frequencies of Table 3, and evaluate the Arf invariants class by class, in particular the sign $-q^7$ on the class of 5-cycles, which carries the leading term. This would make $X=q^6+q^5$ a theorem for odd k and Theorem 7.1 unconditional for odd k. The even-k case, where the orbit invariant j is not complete, is open for both A_6 and X.

Open Problem 9.2. (*the constant Y*) Prove $m_k=O(q^{3/2})$, or at least $m_k=o(q^{5/2})$, which gives $C_6=\lim \mathrm{Var}_{\mathcal{G}^2}/q^9=1$. By Proposition 7.9 this is a weight bound for H^j_c, $9\le j\le 15$, of the eight-dimensional complete intersections $X_0=\{e_1=e_3=e_4=e_5=0\}$ and $X_1=\{e_1=e_3=e_5=0,\ e_4=1\}$ in $\mathbb{A}^{12}$, i.e. a Lefschetz-type statement for singular affine complete intersections whose closures meet the hyperplane at infinity in the base $\mathbb{P}(X_0)$ of the cone X_0; the local ingredients are the Swan conductors of the Artin–Schreier sheaf $\mathcal{L}_\psi(e_4)$ along $\mathbb{P}(X_0)$ and the vanishing cycles at the singular loci (coordinates with fewer than five distinct values).

Open Problem 9.3. (*general d*) Determine the subleading σ-twisted Tate terms of the point count $\Psi(1)=\#V_d^{(\sigma)}(\mathbb{F}_q)$, where $V_d=V(e_1,\dots,e_{d-1})\subset\mathbb{A}^{2d}$: for $d=4$ they are $-q^4+q^3$, for $d=6$ and odd k they are $q^5+q^4-2q^3$ (Theorem 7.1), and for $d=8$ the exact values $\varphi(1)=-40,-6336,-491008$ at $q=2,4,8$ are of size $-1.5\,q^6$ and $-1.9\,q^6$ for $q=4,8$, i.e. at the threshold $q^{3d/4}$. Decide Conjecture 9.2; in particular whether the amplification can reappear for larger d when d is not a power of 2 and $|K_d|=q$.

Note added. Open Problem 9.1 has since been taken up in Parts III and IV [6, 7]. The sextic asked for is $B_\lambda(u)=\lambda_5^2u^6+\lambda_3^2u^2+\lambda_5u+1$: the identity $L'_\lambda=x^4B_\lambda^2+B_\lambda+1$ shows that $\ker L'_\lambda$ is the set of pairwise sums of its six roots, which explains the action of the Frobenius through $S_6\cong Sp_4(\mathbb{F}_2)$ and identifies the class of λ with the factorisation type of B_λ over $\mathbb{F}_q$ (Part III). The sums W' are Frobenius traces of rank-four Witt–Artin–Schreier sheaves; their values are now proved in all eleven classes for odd k (the value $-q^7$ on the class of 5-cycles in Part III, the remaining signs in Part IV through a sixteen-point formula on the coset $T'^{-1}_\lambda(1)$ and a trace-parity theorem). The closed form $X=q^6+q^5$ for odd k therefore depends only on

the polynomial class counts of Table 3, which remain conjectural (verified for $q \leq 512$). Open Problems 9.2 and 9.3 are untouched.

9.4. **Final checklist.** Table 7 lists every statement of the paper with its status (proved, conditional, or open) and the range in which it was checked by exact computation.

TABLE 7. Final checklist: status of the main identities. "exact" means verified by exact integer computation; the method is indicated.

Statement	Status	Range checked
$d = 4$: Theorem 3.2 (all classes)	proved	formula = FFT class by class, $q \le 128$
$d = 4$: $\mathrm{Var}_{\mathcal{G}^2}/q^5 = q(q-1)$, $\mathrm{Var}_{\mathcal{G}}/q^5 = 3 - 3/q$	proved	Lemma 3.10 also checked to $q = 2^{22}$
$d = 4$: ρ structure (Theorem 3.12)	proved	matches [5, Table 4]
$d = 6$: three-value profile (Theorem 4.2)	proved	all q^2 classes, $q \le 32$
$d = 6$: tower identities and reductions	proved	$\mathbb{F}_{2^{12}}$ exhaustive; $q = 2, 4, 8$
$d = 6$: pure cubic and pure quintic parts (Lemmas 6.2, 6.4)	proved, all k	exact, $q \le 128$
$d = 6$: mixed part, k odd (Theorem 6.12); types of $v^5 + v + c$ (Proposition 6.8)	proved	quadratic forms $k \le 7$; L-polynomials $k \le 9$; types $k \le 11$
$d = 6$: $A_6 = q^9 + (q-1)q^5$, k odd (Corollary 6.13)	proved	exact, $q \le 128$ (Table 1)
$d = 6$: X, k odd: structure of W' (Prop. 6.14), $W'(0,0)$, pure cubic vanishing pattern, pure quintic kernel (Lemmas 6.15–6.17)	proved	exact, $q \le 128$
$d = 6$: $X = q^6 + q^5$, k odd (Conjecture 6.19; S_6-structure of Table 3)	open	exact, $q \le 128$; table $k \le 7$
$d = 6$: A_6 and X for even k (Conjecture 6.5)	open	exact, $q \le 128$ (Table 1)
$d = 6$: $\mathrm{Var}_{\mathcal{G}^2}/q^7$ formula (Theorem 7.1)	proved given Conj. 6.5 (X-part only for odd k)	$q \le 32$
$Y = N_0 - N_1$, lower vanishing $H_c^j(X, \mathcal{L}) = 0$ $(j < 9)$, spectral constraints (Props. 7.9, 7.10)	proved	$k \le 5$
$C_6 = 1$	needs $m_k = o(q^{5/2})$, Prop. 7.9(a)	$q \le 32$
spectral folding: $\mathrm{Var}_{\mathcal{G}^2} = \sum_\psi \lvert f_\psi \rvert^2$, fibre sums at $d = 4$ (Theorem 8.2) and $d = 6$, (10)	proved	$d = 4$: $q \le 32$; $d = 6$: $q \le 16$
general even d: translation orbits, K_d by Lucas, lower bound (Proposition 9.1)	proved	$d \le 10$ (Table 6)
general even d: q^2-amplification only for $d = 4, 6$; bounded ratio at $d = 8$ (Conjecture 9.2)	open	$d = 8$: $q \le 8$; $d = 10$: $q \le 4$

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